



Project Report On

Enabling Handover and Energy Efficiency in UAV Networks

*Submitted in partial fulfillment of the requirements for the
award of the degree of*

Bachelor of Technology

in

Computer Science and Engineering

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March 2026

CERTIFICATE

*This is to certify that the project report entitled “**Enabling Handover and Energy Efficiency in UAV Networks**” is a bonafide record of the work done by **Neha Liz Jacob (U2203153)**, **Nitha Theres George (U2203165)**, **Reesa Teddy (U2203174)**, **Visakh V Gireesh (U2203210)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in “Computer Science and Engineering” during the academic year 2025-2026.*

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ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Rev. Dr. Jaison Paul Mulerikkal** CMI, Principal of RSET, and **Dr. Preetha K.G**, Head of the Department of Computer Science and Engineering, for providing us with the opportunity to undertake our project, "Enabling Handover and Energy Efficiency in UAV Networks".

We are highly indebted to our project coordinators, **Ms. Jisha Mary Jose**, Assistant Professor, Dept. of CSE, for their valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express my sincere gratitude to our project guide **Dr. Tripti C** for her patience and all the priceless advice and wisdom she has shared with us.

We would also like to thank our co-guide, **Dr. Moumita Patra**, Associate Professor, IIT Guwahati, for her guidance and encouragement.

Last but not the least, We would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

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Abstract

Maintaining reliable communication in Flying Ad-Hoc Networks (FANETs) remains a significant challenge due to rapid topology changes, fluctuating link quality, and frequent handovers among highly mobile drones. In mission-critical applications such as surveillance, disaster response, and environmental monitoring, Quality of Service (QoS) is not merely a performance metric, it is an operational necessity. Degraded QoS directly translates to missed surveillance frames, delayed emergency commands, and failed sensor data delivery, any of which can compromise the entire mission. Yet conventional signal-strength-based handover strategies cause unnecessary switching, increased latency, packet loss, and uneven load distribution across ground base stations (BSs), producing exactly these outcomes under congested network conditions.

This project proposes a relay-assisted, load-aware handover management framework implemented in Network Simulator-3 (ns-3). The system models 30 drones communicating with three ground BSs under IEEE 802.11ac at 5 GHz. The decision mechanism follows a multi-phase policy: a safety check disconnects drones below receiver sensitivity (-95 dBm), followed by signal-based handovers (triggered below -78 dBm with a 3 dB improvement requirement) and load-based handovers (activated when load imbalance exceeds a threshold after minimum dwell time).

When all candidate BSs are congested, the framework activates a relay-assisted offloading mechanism. A suitable relay drone is selected based on distance, signal quality, and residual capacity, preferably positioned at an optimal altitude to maximize coverage and minimize path loss. The relay maintains the connection and periodically checks for available ground BS capacity, enabling seamless reintegration when resources become available.

Simulation results confirm that the proposed framework preserves service continuity under heavy load by redirecting excess traffic to aerial relays, preventing disconnections and balancing network utilization. The approach achieves up to 15% higher throughput during overload scenarios with a modest delay overhead of 4–8%. Notably, this delay

overhead narrows significantly as network density increases from 25% at low drone counts to just 6.3% at peak density, demonstrating that the framework becomes increasingly efficient under the congested conditions it is designed for.

Overall, integrating load-aware decision making with relay-assisted offloading at optimal UAV positions significantly enhances network reliability, scalability, and Quality of Service in dense FANET deployments, establishing it as a robust solution for next-generation UAV communication infrastructure.

Keywords: Flying Ad-Hoc Networks (FANET), UAV Communication, Handover Management, Load Balancing, Quality of Service (QoS), Network Simulator-3 (ns-3), IEEE 802.11ac, Drone Networks, Wireless Communication Optimization

Contents

Acknowledgment	i
List of Abbreviations	vii
List of Figures	ix
List of Tables	x
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	2
1.3 Scope and Motivation	2
1.4 Objectives	2
1.5 Challenges	3
1.6 Assumptions	3
1.7 Societal / Industrial Relevance	3
1.8 Organization of the Report	4
1.9 Summary of Chapter	4
2 Literature Survey	5
2.1 Existing Methods	5
2.1.1 The Evolution of Networks and Management in a 6G World: An Inventor's View [1]	5
2.1.2 Survey of Important Issues in UAV Communication Networks [2] . .	7
2.1.3 Towards Smart 6G: Mobility Prediction for Dynamic Edge Services Migration [3]	10
2.1.4 Proactive Handover Decision for UAVs with Deep Reinforcement Learning [4]	12

2.1.5	Spatial Disparity of QoS Metrics Between Base Stations in Wireless Cellular Networks [5]	15
2.1.6	Explainable AI for UAV Mobility Management: A Deep Q-Network Approach for Handover Minimization [6]	18
2.1.7	Altitude Optimization and Task Allocation of UAV-Assisted MEC Communication System [7]	22
2.1.8	Optimal LAP Altitude for Maximum Coverage [8]	25
2.2	Summary and Gaps Identified	29
2.3	Summary of Chapter	31
3	System Design	32
3.1	Architecture Overview	32
3.1.1	Air Layer	32
3.1.2	Ground Layer	33
3.1.3	Base Station States	33
3.1.4	Decision Engine	36
3.1.5	System Workflow	37
3.1.6	Key Architectural Features	38
3.2	Component Design	39
3.3	Sequence Diagram	40
3.3.1	Phase 1: Measure	41
3.3.2	Phase 2: Decide	41
3.3.3	Phase 3: Connect	41
3.3.4	Phase 4: Disconnect	41
3.3.5	Phase 5: Complete	42
3.4	Module Division	42
3.4.1	Module 1: Drone and Base Station Environment	42
3.4.2	Module 2: Drone Mobility and Handover Decision Engine	42
3.4.3	Module 3: Height-Based Communication and Cooperative Offloading	44
3.4.4	Module 4: GUI and Visualization	44
3.5	Tools and Technologies	44
3.6	Work breakdown	45

3.7	Key Deliverables	46
3.8	Project Timeline	46
3.9	Summary of chapter	47
4	Proposed Method	48
4.1	System Description	48
4.2	Priority-Based Handover Decision Algorithm	48
4.3	Summary of Chapter	53
5	Results and Discussions	54
5.1	Throughput Analysis	54
5.2	Delay Analysis	55
5.3	Performance Comparison Table	57
5.4	Discussion	57
5.5	Summary of Chapter	58
6	Conclusion and Future Work	59
6.1	Conclusion	59
6.2	Future Work	59
6.3	Summary	60
	References	61
	List of Publications	63
	Appendix A: Presentation	64
	Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes	87
	Appendix C: CO-PO-PSO Mapping	92

List of Abbreviations

A2G	Air-to-Ground
AI	Artificial Intelligence
BS	Base Station
C++	C Plus Plus (Programming Language)
DL	Downlink
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
GUI	Graphical User Interface
HAP	High-Altitude Platform
HO	Handover
IoT	Internet of Things
IIT	Indian Institute of Technology
LAP	Low-Altitude Platform
LoS	Line-of-Sight
MEC	Mobile Edge Computing
ML	Machine Learning
NLoS	Non-Line-of-Sight
NS-3	Network Simulator 3
PPO	Proximal Policy Optimization
QoS	Quality of Service
RAM	Random Access Memory
RL	Reinforcement Learning
RSSI	Received Signal Strength Indicator
RX	Receiver

SHAP	SHapley Additive exPlanations
UAV	Unmanned Aerial Vehicle
UAV-BS	UAV Base Station (Relay Node)
UE	User Equipment
UL	Uplink
6G	Sixth Generation (Wireless Network)

List of Figures

3.1	Proposed System Architecture for UAV Handover and Energy Efficiency	38
3.2	Dataflow Diagram	39
3.3	Sequence Diagram	40
3.4	Gantt Chart	47
5.1	Throughput vs. Number of Drones	55
5.2	Average Delay vs. Number of Drones	56

List of Tables

2.1	Comparative Summary of Literature Survey	29
3.1	Base Station State Transition Summary	35
4.1	Handover Decision Algorithm Parameters	50
5.1	Performance Comparison — Cooperative Offloading vs. Load-Based Handover . .	57

Chapter 1

Introduction

This chapter introduces the concept of the project “Enabling Handover and Energy Efficiency in UAV Networks.” It outlines the motivation behind developing a smarter connectivity management framework for unmanned aerial vehicles (UAVs), the core challenges faced in aerial mobility, and the technological approaches adopted to address them. The subsequent sections describe the problem context, system objectives, and the overall relevance of this work in the evolving landscape of intelligent wireless communication..

1.1 Background

The rapid advancement of Unmanned Aerial Vehicle (UAV) technology has opened transformative possibilities across domains including military operations, disaster relief, precision agriculture, and smart urban infrastructure management. Central to the effective deployment of these systems is the ability to maintain reliable, low-latency wireless communication between UAVs and ground-based infrastructure as the vehicles move dynamically across three-dimensional airspace. In conventional cellular networks, handover transferring an active connection from one base station to another is well-studied through signal-strength thresholds and load-balancing policies. However, UAVs introduce fundamentally different challenges. Unlike terrestrial users constrained to two-dimensional movement, UAVs operate in 3D space, experiencing rapid changes in altitude, speed, and trajectory that cause frequent signal fluctuations, rendering standard handover algorithms unreliable and inefficient. A particularly disruptive phenomenon is the ping-pong effect, wherein a drone repeatedly switches association between two nearby base stations due to minor signal fluctuations at cell boundaries. Each unnecessary handover introduces signaling overhead, momentary connectivity interruptions, and increased energy consumption highly undesirable in mission-critical UAV applications. Furthermore, lower-altitude drones frequently encounter Non-Line-of-Sight (NLoS) propagation conditions due to terrain obstructions, while higher-altitude drones benefit from clearer LoS links to ground base stations. Energy constraints compound these challenges. Battery capacity limits UAV operational du-

ration, making it essential that communication mechanisms including handover procedures be optimized for energy efficiency. Wasteful ping-pong handovers and redundant relay operations must be minimized to extend the operational lifetime of the drone fleet.

1.2 Problem Definition

UAVs operating in 3D environments face frequent disconnections due to rapid signal fluctuations that conventional RSSI-based handover mechanisms cannot handle, making an adaptive multi-criteria approach one that considers signal quality, network load, and altitude-dependent LoS conditions essential for reliable connectivity.

1.3 Scope and Motivation

The scope of this project encompasses the design, implementation, and simulation-based evaluation of an adaptive handover and cooperative offloading framework for UAV communication networks. The system is modeled within a 2000m x 2000m simulation environment with 30 UAVs and three ground base stations. The handover engine incorporates three prioritized modes alongside a height-aware offloading module grounded in the Air-to-Ground path loss model.

The motivation stems from the growing deployment of UAV-assisted networks in time-critical applications where communication failures are unacceptable. UAVs operating in three-dimensional environments experience frequent handovers due to their dynamic mobility patterns, resulting in connectivity disruptions that undermine network reliability. Reliable, low-latency communication is nonetheless essential for critical UAV applications such as surveillance and disaster management, where service interruptions are unacceptable. Compounding this challenge, the inherent energy constraints of UAV platforms demand that handover mechanisms be power-efficient, minimizing unnecessary switching and associated battery drain. Existing approaches such as RSSI-based handover methods fall short in this regard, as they lack the adaptivity and mission-awareness needed to handle the complex, altitude-dependent communication conditions characteristic of real-world UAV deployments.

1.4 Objectives

- To improve handover performance in UAV communication networks through adaptive, multi-criteria decision mechanisms.

- To develop a priority-based handover algorithm combining urgent, signal-based, and load-based handover triggers for 3D UAV mobility.
- To implement a height-based cooperative offloading strategy using the Air-to-Ground path loss model to maximize LoS communication probability.
- To ensure reliable connectivity between UAVs and base stations, minimizing disconnections during flight.
- To evaluate the throughput-latency trade-off of cooperative relay-based offloading under varying network load conditions.

1.5 Challenges

Even with advanced systems, several real-world challenges continue to hinder the reliable deployment of UAV communication networks. Battery life remains a critical constraint, as continuous monitoring and video transmission rapidly drain drone batteries, limiting operational duration and necessitating frequent recharging or replacement. Unpredictable environmental circumstances such as birds, adverse weather conditions, and sudden physical obstacles can disturb a drone's flight path and introduce abrupt signal variations that challenge handover stability. Signal interference arising from overlapping frequencies and environmental obstructions further reduces network reliability, degrading the quality of communication links between drones and base stations. Finally, coordinating multiple drones simultaneously demands precise timing and collision avoidance strategies to prevent inter-drone interference, adding significant complexity to network management in dense UAV deployments.

1.6 Assumptions

The following assumptions are made in the design and evaluation of the proposed system:

1. Each Base Station (BS) can support a maximum of 10 drones simultaneously.
2. Each drone transmits one data packet per time tick to the network.
3. A drone maintains only a single base station connection at any given time.

1.7 Societal / Industrial Relevance

The proposed framework has direct applicability in military surveillance, where continuous UAV-to-base communication is critical for real-time situational awareness. In disaster management,

UAV swarms with intelligent handover mechanisms can maintain data links in degraded infrastructure environments, facilitating search-and-rescue operations. Smart city UAV deployments for traffic monitoring and emergency response also benefit from the reliable handover guarantees offered by the framework. Industrially, the work contributes to the foundational communication stack required for UAV integration in 6G networks.

1.8 Organization of the Report

Chapter 1 provides an overview of the project, including background, problem definition, scope, objectives, challenges, assumptions, and societal relevance. Chapter 2 presents a survey of related work on UAV handover, mobility management, altitude optimization, and cooperative offloading. Chapter 3 describes the system design, architecture, and module decomposition. Chapter 4 provides a detailed technical description of the implementation. Chapter 5 presents and analyzes simulation results. Chapter 6 concludes the report and outlines future work directions.

1.9 Summary of Chapter

This chapter introduced the motivating problem of enabling reliable handover and energy-efficient communication in UAV networks. The limitations of conventional approaches in 3D dynamic environments were discussed, and the scope, objectives, and contributions of the proposed framework were outlined.

Chapter 2

Literature Survey

This chapter reviews the existing body of work related to UAV communication networks, handover mechanisms, mobility prediction, altitude optimization, and cooperative offloading. Eight key papers are surveyed, and a comparative analysis identifies the gaps that motivate the proposed work.

2.1 Existing Methods

2.1.1 The Evolution of Networks and Management in a 6G World: An Inventor's View [1]

Problem Addressed:

- Networks evolving toward 6G must handle ultra-dense connectivity, AI-native operations, and seamless integration of physical and digital systems.
- Traditional network management frameworks are static, human-dependent, and lack the intelligence and automation required for next-generation systems.
- Complexity in managing large-scale distributed resources, service orchestration, and dynamic user demands makes manual management inefficient and prone to delays.
- Limited understanding exists of how AI, intent-based networking, and self-optimizing mechanisms will reshape the management architecture of future 6G networks.

Objective:

- Provide an inventor's perspective on the evolution from 5G to 6G, focusing on network management and automation.
- Identify architectural shifts required to support intelligent, self-managed, AI-driven 6G networks.

- Propose a conceptual framework integrating zero-touch management, intent-based orchestration, and AI-enabled decision systems.
- Highlight challenges and opportunities for sustainable, adaptive, and user-centric 6G ecosystems.

Methodology:

1. Conceptual Analysis

Review of network evolution trends, existing management systems, and future use cases. Identification of technological enablers such as AI, edge intelligence, and semantic communication.

2. Framework Design:

Three-layered management model:

- *Intent Layer*: Captures user or business goals in high-level semantics.
- *Automation Layer*: Interprets intents and optimizes resource allocation using AI and knowledge graphs.
- *Execution Layer*: Applies configurations autonomously via programmable interfaces and real-time feedback loops.

3. AI and Cognitive Management:

Cognitive agents continuously learn from data across network slices, edge devices, and service platforms. Reinforcement learning and self-supervised models enable adaptive optimization.

4. 6G Integration Perspective:

Discusses integration with digital twins, terahertz communication, and distributed cloud-native infrastructures. Emphasizes proactive prediction and self-healing capabilities.

Vision and Implications:

- 6G envisioned as an AI-native network with largely automated and self-optimizing management.
- Principles include zero-touch, intent-based, and self-healing management.
- Shift from device-centric to service-centric management, prioritizing Quality of Experience (QoE).

- Conceptual groundwork for fully autonomous network ecosystems integrating sustainability, scalability, and intelligence.

Integration with 6G Communication Trends:

- Future 6G networks will enable digital twin worlds, integrating physical, biological, and digital environments.
- AI-driven air interfaces and cognitive spectrum sharing complement autonomous management and resource allocation.
- RAN-Core convergence, enhanced localization, sensing, and extreme performance requirements support self-optimizing networks.
- Security and privacy are foundational, aligning with zero-touch, AI-native network operations.

Conclusion:

This section shows that the evolution of 6G networks combines advanced management paradigms (zero-touch, intent-based, AI-driven) with technological innovations (digital twins, AI air interface, terahertz communication) to create autonomous, high-performance, and user-centric wireless systems.

2.1.2 Survey of Important Issues in UAV Communication Networks [2]

Problem Addressed:

- UAV networks exhibit fluid topology, intermittent links, and high node mobility in both 2D and 3D space, making them fundamentally different from MANETs and VANETs.
- Existing routing protocols designed for MANETs and VANETs do not fully address the unique characteristics of UAV networks, including rapid topology changes, frequent node removal and addition, and strict power constraints.
- Seamless handover in UAV networks remains an open research problem, as traditional mobility management protocols suffer from high latency, packet loss, and poor scalability under high-mobility aerial conditions.
- Energy constraints are significantly more severe in small UAVs compared to vehicular or mobile ad hoc networks, with typical flight times of only 25–30 minutes, making energy-efficient protocol design critical.

Objective:

- To survey the state of the art in UAV communication networks, focusing specifically on the understudied areas of self-organization, routing, seamless handover, and energy efficiency.
- To categorize UAV networks based on application type, mobility characteristics, and network architecture, distinguishing them from MANETs and VANETs.
- To evaluate the applicability of existing routing protocols including static, proactive, reactive, hybrid, and geographic 3D protocols to UAV network environments.
- To analyze delay and disruption tolerant networking strategies suitable for UAV networks prone to frequent partitioning.
- To review energy conservation techniques across the physical, data link, and network layers, and identify cross-layer approaches applicable to UAV deployments.

Methodology:**1. Network Characterization:**

UAV networks are categorized based on application domain (Internet delivery, sensing, attack), node mobility speed, network architecture (star vs. mesh), and control model (centralized vs. distributed). Key distinctions from MANETs and VANETs are identified, including 3D mobility, energy constraints, and fluid topology.

2. Routing Protocol Analysis:

Four classes of routing protocols are evaluated for applicability to UAV networks:

- *Static protocols*: Fixed routing tables; not fault-tolerant; limited applicability.
- *Proactive protocols (OLSR, DSDV, BABEL, B.A.T.M.A.N.)*: Maintain up-to-date tables but incur high overhead unsuitable for bandwidth-constrained UAV networks.
- *Reactive protocols (DSR, AODV)*: On-demand but suffer from route discovery delays and throughput degradation under intermittent links.
- *Hybrid protocols (ZRP, TORA)*: Balance overhead and latency but add complexity; suitability for UAV networks is limited.

3. Delay and Disruption Tolerant Networking (DTN):

Store-carry-forward (SCF) architecture is identified as the appropriate basis for UAV networks prone to partitioning. Routing strategies reviewed include:

- Deterministic routing: Applicable when UAV trajectories are known in advance.
- Epidemic routing and Spray-and-Wait: Stochastic flooding approaches; effective but resource-intensive.
- Estimation-based routing (PROPHET): Uses historical encounter probability for forwarding decisions.
- Node-movement-control and coding-based approaches: Controlled ferrying and erasure coding for improved delivery in partitioned networks.

4. Seamless Handover Analysis:

Handover types (hard/soft, horizontal/vertical) are examined in the context of UAV networks. Existing mobility management protocols MIPv4, MIPv6, PMIPv6, HMIPv6, FMIPv6, and NEMO are evaluated for UAV applicability. IEEE 802.21 Media Independent Handover (MIH) is identified as a promising but nascent standard for heterogeneous UAV handover management.

5. Energy Efficiency Survey:

Energy conservation techniques are reviewed across three protocol layers:

- Network layer: Path selection (EMM-DSR, FAR, Minimum-energy routing), node selection (LEAR, DEAR), coordinator-based (CBRP, GAF, Span), and sleep-based (CDS, EAR) approaches.
- Data link layer: Duty cycle protocols with single and dual radios (S-MAC, T-MAC, STEM, PAMAS), topology/power control (XTC, LFTC), and cluster-based MAC protocols (LEACH, PEGASIS).
- Physical layer: Dynamic voltage scaling (DVS), node-level power scheduling (LM-SPT), minimum subset selection, and sleep schedules with buffering (SRA).
- Cross-layer protocols: MTEC+ACW and CLEEP for joint optimization across layers.

6. SDN for UAV Network Automation:

Software Defined Networking (SDN) using OpenFlow is examined as a mechanism to automate UAV network control, enabling flexible routing, interference reduction, and dynamic reconfiguration in response to node failures and topology changes.

Key Findings and Outcomes:

- No existing routing protocol is natively suited to UAV networks; all require modifications or replacement to handle 3D mobility, intermittent links, and energy constraints[9].
- Mesh network architecture with self-organizing and self-healing capabilities is identified as the most suitable topology for multi-UAV deployments.
- Seamless handover in UAV environments remains largely unsolved; MIH offers a path forward but requires broader deployment and evaluation.
- Energy consumption by communication equipment can reduce UAV flight time by 20% or more, making cross-layer energy optimization a critical research priority.
- SDN provides a promising programmable control framework for UAV networks, though its application to infrastructure-less wireless environments requires further investigation.

Limitations:

- The survey predates developments in machine learning-based handover management, 5G/6G aerial user equipment integration, and altitude-aware Air-to-Ground propagation models[10].
- Handover mechanisms are discussed conceptually without proposing concrete algorithms or simulation results specific to UAV 3D mobility.
- Energy models do not account for the combined effect of propulsion and communication power consumption, which is critical for accurate operational lifetime estimation.

2.1.3 Towards Smart 6G: Mobility Prediction for Dynamic Edge Services Migration [3]

Problem Addressed:

- With the increasing mobility of users, drones, and connected vehicles, seamless service continuity in edge-assisted 6G networks has become a major challenge.
- Frequent handovers and unpredictable mobility patterns often lead to latency, service interruptions, and inefficient resource utilization during service migration between edge servers.
- Existing service migration techniques are typically reactive, triggering migrations only after performance degradation occurs.

- There is a lack of efficient predictive mechanisms capable of forecasting user mobility and performing proactive edge service migration to maintain high Quality of Service (QoS).
- Prior studies in cellular networks highlight the benefits of LSTM-based mobility prediction: [8] showed that predicting user trajectories reduces handover failures and improves network reliability, while [5] demonstrated improved handover accuracy and resource management using LSTM networks. Additionally, predictive handover mechanisms in 5G networks such as Deep-Mobility have shown that deep learning can significantly enhance reliability and reduce service disruption, motivating similar approaches for edge service migration in 6G.

Objective:

- Design a predictive framework that anticipates user mobility patterns to enable proactive edge service migration in 6G networks.
- Reduce latency, minimize migration overhead, and ensure continuous connectivity during user movement.
- Integrate AI-driven prediction models into the edge management architecture for dynamic and intelligent decision-making, leveraging LSTM-based techniques validated in prior cellular handover studies and predictive 5G handover studies .
- Enhance network adaptability by aligning migration timing with mobility forecasts, ensuring high Quality of Experience (QoE) for users.

Methodology:

1. System Architecture:

The proposed 6G edge architecture integrates distributed edge servers, mobile users (UEs), and a centralized controller for mobility management. Each edge node hosts virtualized services that can migrate dynamically based on predicted user trajectories.

2. Mobility Prediction Module:

An LSTM-based machine learning model is trained using user movement data including speed, direction, cell ID transitions, and dwell time . The model predicts the next probable edge zone or cell region where the user will move, employing time-series forecasting and spatial-temporal feature extraction to improve prediction accuracy.

3. Proactive Migration Decision:

Once the next location is predicted, the edge controller proactively prepares the target node for migration. Service state is pre-transferred before the actual handover occurs, significantly reducing service downtime and extending the advantages shown in LSTM-based handover prediction studies [9] and deep learning-based 5G handover studies [11] to edge service migration.

4. Evaluation and Metrics:

The framework is evaluated through simulations measuring metrics such as latency, handover rate, and migration cost. Compared against traditional reactive approaches, the system demonstrates higher reliability, reduced migration delay, and superior accuracy in mobility forecasting, echoing the improvements observed in predictive 5G handover mechanisms .

Conclusion:

- The proposed AI-based predictive mechanism effectively reduces service interruption and latency during handovers.
- Demonstrates superior accuracy in mobility forecasting compared to conventional mobility models, leveraging the proven benefits of LSTM-based handover prediction and predictive 5G handover approaches .
- Enhances Quality of Experience (QoE) by ensuring that edge services are pre-deployed near users' predicted locations.
- Provides a strong foundation for context-aware, self-optimizing, and proactive management in 6G networks, bridging cellular mobility prediction research with next-generation edge service orchestration.

2.1.4 Proactive Handover Decision for UAVs with Deep Reinforcement Learning [4]

Problem Addressed:

- UAVs deployed in critical applications such as aerial surveillance, logistics, real-time monitoring, and disaster management demand continuous, reliable connectivity to cellular networks for command, control, and data transmission.

- Existing cellular handover schemes are primarily designed for ground-based users and struggle to cope with the unique mobility characteristics of UAVs, including rapid and unpredictable 3D movements at varying altitudes, frequent signal fluctuations due to line-of-sight propagation and interference, and the need to maintain low-latency links across multiple overlapping base station coverage zones.
- In traditional systems, UAVs often trigger excessive or unnecessary handovers due to high visibility to multiple base stations simultaneously, resulting in ping-pong effects, degraded throughput, and in some cases connection drops that directly impact mission success.
- Huang et al. modeled the 3D UAV network structure analytically using stochastic geometry to estimate handover probability, while Lin et al. validated these challenges through real-world LTE drone flight trials, demonstrating that aerial users experience higher interference and more frequent handovers than ground users.

Objective:

- Design a proactive, intelligent, and adaptive handover decision framework for UAVs using Deep Reinforcement Learning (DRL).
- Minimize redundant or unnecessary handovers while maintaining stable and high-quality connectivity under dynamic flight conditions.
- Adapt automatically to diverse network topologies and UAV trajectories without retraining from scratch.
- Offer a scalable learning-based approach that generalizes across different deployment environments.

Methodology:

1. State Representation:

The UAV's environment is represented through a multi-dimensional state vector that includes 3D position, velocity, and altitude; current serving BS and neighboring BS signal strengths (RSSI); interference metrics observed at different heights; and environmental factors such as urban obstacles or dynamic channel variations.

2. Action Space:

The DRL agent can choose between two primary actions: perform a handover by switching to a new BS predicted to provide better long-term connectivity, or stay connected to

the current BS if stability outweighs short-term RSSI gains. Unlike conventional reactive algorithms that trigger handovers only after a signal drop, this framework makes proactive, predictive decisions based on movement and network trends.

3. Reward Function Design:

The reward structure integrates network-level metrics inspired by the empirical results and the probabilistic handover modeling . Positive rewards are assigned for maintaining strong RSSI, reducing delay, and sustaining stable links, while negative rewards are applied for abrupt or unnecessary handovers, high latency, or unstable connectivity. This design encourages the agent to balance signal quality and handover efficiency.

4. Learning Algorithm:

The model employs Proximal Policy Optimization (PPO), a robust DRL algorithm ideal for continuous, high-dimensional state spaces and real-time UAV decision-making. PPO updates the policy iteratively based on cumulative rewards, prevents instability through clipped policy updates, and ensures stable convergence even as UAV dynamics evolve.

5. Simulation and Validation:

A 3D UAV network simulation environment is developed reflecting the realistic scenarios and altitude-based effects are observed . The simulation models multiple base stations with varying coverage radii and interference levels, UAV trajectories generated using mobility models derived from [5]’s stochastic geometry framework, and dynamic network conditions including traffic density and environmental noise.

6. Memory and Adaptation:

Through DRL-based learning, the UAV autonomously refines its decision-making policy by effectively remembering previous optimal actions. The system adapts to new environments, UAV speeds, or flight altitudes without retraining from scratch, making it suitable for large-scale deployment in real-world networks.

Performance and Outcomes:

- Reduces unnecessary handovers by up to 70–75% compared to traditional reactive and Q-learning methods.
- Maintains stable RSSI above -75 dBm over 80% of flight time.
- Improves spectral efficiency by minimizing redundant signaling overhead.

- Ensures seamless connectivity during high-speed UAV movement across diverse 3D aerial environments.

Limitations:

- The framework does not fully model altitude-dependent signal variations specific to UAV operations, particularly the Air-to-Ground path loss characteristics that govern LoS probability at different heights.
- The computational overhead of PPO training limits real-time applicability on resource-constrained UAV platforms with limited battery and processing capacity.
- Cooperative relay offloading as a fallback mechanism for blocked handovers is not considered, leaving drones without recourse when all candidate base stations are at full capacity.
- Generalization across highly heterogeneous network topologies with varying base station densities and coverage configurations remains an open challenge.

2.1.5 Spatial Disparity of QoS Metrics Between Base Stations in Wireless Cellular Networks [5]

Problem Addressed:

- A significant spatial disparity of QoS metrics including cell load, users' number, and user throughput exists between base stations in large cellular networks due to the irregularity of cell geometry, varying BS deployment environments, and inter-cell interference dependencies.
- Base stations are located in heterogeneous environments (urban, suburban, and rural) with different heights, transmission powers, and propagation characteristics, causing QoS metrics to vary substantially across cells without obvious dependence on local traffic demand.
- Existing approaches rely on time-consuming typical cell simulations to estimate QoS distributions, making them impractical for network dimensioning tasks such as bandwidth planning or BS density optimization.
- Prior works do not provide explicit closed-form expressions for the spatial cumulative distribution function (CDF) of QoS metrics such as cell load, users' number, and user throughput that account for the dynamics of call arrivals and departures.

Objective:

- Develop a fully analytical framework to explicitly characterize the spatial disparity of QoS metrics cell load, users' number, and user throughput across base stations in large heterogeneous wireless cellular networks.
- Build explicit CDF expressions for these metrics using stochastic geometric and queueing theoretic techniques, taking into account random BS locations modeled as a Poisson point process with log-normal shadowing.
- Propose a computationally efficient *mean cell model* that approximates the averages, variances, and distributions of QoS metrics without requiring full typical cell simulations.
- Validate the proposed analytical expressions against both simulation results and real-field measurements from operational 3G and 4G networks.

Methodology:**1. Network Model:**

BS locations are modeled as a stationary, simple, and ergodic Poisson point process on R^2 with intensity λ . Each BS emits a random transmission power, and propagation loss incorporates both a deterministic distance-dependent path loss component and a log-normal shadowing process. The downlink SINR at any user location is expressed as:

$$\text{SINR}(y, \Phi) = \frac{1/L_X(y)}{N + \sum_{Y \in \Phi \setminus \{X\}} \varphi_Y / L_Y(y)},$$

where interference factors $\varphi_Y = \min(\theta(Y), 1)$ reflect the probability that a BS is actively serving users.

2. Local QoS Metrics via Queueing Theory:

Each BS is modeled as a multi-class processor sharing queue. The cell load, mean number of users, and mean user throughput are expressed analytically as:

$$\begin{aligned} \theta(X) &= \rho_d \int_{\tilde{V}(X)} \frac{1}{R(\text{SINR}(y, \Phi))} dy, \\ N(X) &= \frac{\theta(X)}{1 - \theta(X)}, \quad \theta(X) < 1, \\ r(X) &= \max(\rho_c(X) - \rho(X), 0). \end{aligned}$$

The cell loads across all BSs are coupled through a system of nonlinear *cell-load equations*, whose unique solution is computed iteratively.

3. Mean Cell Model:

A simplified virtual *mean cell* is introduced, characterized by a mean traffic demand $\tilde{\rho} = \rho_d/\lambda$ and a mean load $\tilde{\theta}$ obtained by solving a single fixed-point equation. This avoids the need to solve the full system of coupled load equations and provides a computationally efficient approximation of network-wide QoS averages.

4. QoS Distribution Derivation:

Empirical observations confirm that the distributions of traffic demand and cell load are approximately log-normal. Leveraging this observation, explicit CDF expressions are derived:

- Traffic CDF: Characterized by its mean $\bar{\rho}$ and variance $v_\rho = \alpha_\rho \bar{\rho}^2$, where α_ρ is a network-invariant traffic slope constant.
- Load CDF: Characterized by its mean $\bar{\theta}$ and variance $v_\theta = \alpha_\theta \bar{\theta}^2$, where α_θ is a load slope constant validated numerically.
- Users' Number CDF: Derived from the load CDF using the queueing relation $N = \theta/(1 - \theta)$.
- User Throughput CDF: Derived from the joint log-normal distribution of traffic demand and critical traffic using the relation $r = \max(\rho_c - \rho, 0)$.

5. Validation:

The proposed analytical expressions are validated by comparing them against typical cell model simulations and real-field measurements collected from an operational network covering a large heterogeneous region of a European country, for both 3G (2.1 GHz, 5 MHz bandwidth) and 4G (2.6 GHz, 20 MHz bandwidth) deployments across 24 hours of network operation.

6. Bandwidth Dimensioning Application:

The derived user throughput CDF is applied to solve the frequency bandwidth dimensioning problem determining the minimum bandwidth required to ensure that 95% of cells offer a user throughput above a target threshold. The analytical approach reduces computation time from approximately 46 days (via simulation) to under 5 minutes.

Key Findings and Outcomes:

- The distributions of traffic demand and cell load across BSs are well approximated by log-normal distributions, with variances proportional to the square of their respective

means.

- The proposed mean cell model provides accurate approximations of QoS metric averages and distributions without requiring full system simulations, enabling efficient network dimensioning.
- Analytical CDF expressions for all four QoS metrics show strong agreement with both typical cell simulations and real-field measurements across all traffic conditions where average load $\bar{\theta} \leq 0.7$.
- The framework is directly applicable to heterogeneous multi-tier networks through scaling laws that preserve the relationship between QoS distributions and mean per-cell traffic demand across urban, suburban, and rural zones.

Limitations:

- The framework is developed for downlink performance only; an equivalent model for the uplink is identified as a direction for future work.
- The log-normal approximation for load CDF overestimates loads at high traffic conditions ($\bar{\theta} > 0.7$), limiting accuracy in heavily congested network regions.
- The model assumes users do not move during their sessions, which is a simplification that may not capture the dynamic handover behavior experienced by aerial users such as UAVs operating at varying altitudes and speeds.
- The stochastic geometry framework models BSs as a homogeneous Poisson point process, which may not fully capture the structured, non-random BS placement typical of real-world deployments, particularly in the context of UAV ground base station associations.

2.1.6 Explainable AI for UAV Mobility Management: A Deep Q-Network Approach for Handover Minimization [6]

Problem Addressed:

- The integration of UAVs into cellular networks presents significant mobility management challenges, primarily due to frequent handovers caused by probabilistic line-of-sight conditions with multiple ground base stations.
- Unlike terrestrial users, UAVs experience frequent handovers due to their 3D mobility and the fragmented, overlapping coverage of terrestrial base stations in the sky, disrupting

service continuity and generating excessive control signaling that degrades overall network efficiency.

- Traditional handover decision metrics such as Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) are insufficient for UAVs, as they lead to unnecessary handovers and ping-pong effects due to higher line-of-sight probability with an increased number of candidate base stations.
- Unnecessary handovers cause uplink data loss due to the limited buffer capacity of UAVs, and existing reinforcement learning-based handover approaches function as black boxes, limiting interpretability and trust in safety-critical deployments.
- The opacity of AI-driven decision-making in emerging 6G systems raises serious concerns about trust, validation, and regulatory compliance, making explainability a fundamental requirement for practical deployment.

Objective:

- Propose a DQN-based handover optimization framework that integrates buffer queue status alongside traditional RSRP and RSRQ metrics to enable more adaptive and intelligent handover decisions for cellular-connected UAVs.
- Minimize the number of unnecessary handovers and reduce ping-pong effects while maintaining reliable uplink connectivity and preventing buffer overflow.
- Incorporate Shapley Additive Explanations (SHAP) to quantify the contribution of each state parameter to handover decisions, improving the interpretability and transparency of the learned RL policy.
- Bridge the gap between AI-driven models and human decision-makers by integrating language models to generate natural language explanations for RL-based handover decisions.
- Validate the framework using real-world LTE data from UAV flight trials to demonstrate its practical effectiveness beyond simulation environments.

Methodology:

1. System Model and Problem Formulation:

An uplink communication scenario is considered where K UAVs transmit to N fixed base

stations over orthogonal resource blocks. The 3GPP Direct Air-to-Ground (DA2G) channel model is adopted, differentiating between LoS and NLoS propagation with altitude-dependent path loss models. The buffer queue evolution is modeled using a Poisson packet arrival process, where handover events generate additional control packets that increase buffer occupancy. The optimization objective is to jointly minimize buffer congestion and handover frequency subject to constraints on BS association, maximum buffer occupancy, and minimum RSRP and RSRQ thresholds.

2. DQN-Based Handover Management:

The handover decision problem is formulated as a reinforcement learning task defined by the tuple (S, A, r) . The state space $S_k(t)$ for UAV k at time t encapsulates 3D position coordinates, buffer queue size, serving BS identity, and RSRP and RSRQ measurements from the set of strongest candidate BSs. The action space consists of the BS association decision $\Phi_{k,l}$. The reward function balances four objectives through non-negative weights: reducing buffer occupancy, penalizing handover events, and rewarding maintenance of RSRP and RSRQ above their respective thresholds. A Proximal Policy Optimization-compatible DQN architecture with experience replay, target network stabilization, and an ϵ -greedy exploration policy with decaying learning rate is employed for training.

3. Explainable AI Framework:

The XAI framework comprises three integrated modules. First, the trained DQN policy $\pi(s, \theta)$ serves as the black-box model. Second, SHAP-based feature attribution computes Shapley values $\psi_i(\Phi)$ to quantify the marginal contribution of each state parameter to the BS association decision, using DeepSHAP which combines SHAP with DeepLIFT to efficiently propagate contribution scores backward through the neural network. Third, an interpretability module uses language models to generate human-readable natural language explanations from SHAP values, action logits, input features, and contextual metadata, with predefined templates, confidence scores, and human-in-the-loop validation to mitigate hallucination risks.

4. Evaluation with Real-World Data:

The framework is validated using real-world LTE network performance data collected during UAV flight trials with a commercial DJI-M600 UAV equipped with two LTE modems. The dataset comprises approximately 90 flights recorded in urban and rural settings, including uplink RSRP and RSRQ measurements under operational conditions. The DQN-based approach is compared against a conventional handover management (CHM) scheme

with fixed parameters and the model-based Service Availability-based Mobility Robustness Optimization (SA-MRO) approach.

Key Results:

- The DQN-based approach achieves a reduction of more than 80% in the number of handovers compared to the conventional CHM scheme, limiting the ping-pong rate to approximately 14% versus 40% for CHM and 27% for SA-MRO.
- SHAP analysis reveals that buffer queue size is the most influential feature in both simulated and real-world data, with high buffer values consistently driving positive handover decisions.
- UAV position ranks as the second most important feature, while signal quality metrics RSRQ and RSRP exhibit more scattered and environment-dependent impacts, highlighting the gap between simulated and real-world signal dynamics.
- The natural language explanation module successfully generates intuitive, human-readable rationales for individual handover decisions, effectively bridging the gap between AI-driven models and network operators.

Limitations:

- The framework considers a single centralized agent trained offline; extension to multi-agent reinforcement learning for coordinated mobility management across multiple simultaneous UAVs remains future work.
- The altitude-dependent path loss model assumes $P_{\text{LoS}} = 1$ for UAV altitudes above 100 m, which may not accurately reflect signal conditions in complex urban environments with irregular building distributions.
- Language model integration introduces risks of hallucination and confirmation bias in generated explanations, requiring ongoing human-in-the-loop validation that may not scale to large deployments.
- The cooperative relay offloading mechanism as a fallback when no candidate base station meets the RSRP and RSRQ thresholds is not addressed, a gap directly relevant to the present project's height-aware offloading module.

2.1.7 Altitude Optimization and Task Allocation of UAV-Assisted MEC Communication System [7]

Problem Addressed:

- The growing demand for reliable, high-speed maritime wireless communication driven by marine tourism, oceanic mineral exploration, and maritime rescue cannot be adequately met by traditional satellite or HF/VHF communication systems due to high cost, large transmission delay, and insufficient channel bandwidth.
- Conventional ground-based MEC servers have fixed deployment locations and limited coverage, making them unsuitable for serving mobile maritime users operating in environments where network infrastructure is scarce.
- Existing UAV-assisted MEC research assumes fixed UAV altitude without optimizing the flying altitude, despite altitude being a critical parameter that governs the Air-to-Ground path loss, LoS probability, and ultimately the achievable transmission rate and system delay.
- Prior works on partial task offloading to a single UAV do not account for the limited computational capacity of the UAV, making it impractical for compute-intensive tasks; a cooperative offloading strategy involving both the UAV and an onshore base station is necessary to alleviate computational burden and minimize total system latency.
- No closed-form analytical solution for the jointly optimal UAV flying altitude and task allocation ratio exists in the literature for UAV-assisted MEC systems with probabilistic LoS/NLoS channel models[11].

Objective:

- Propose a UAV-assisted MEC uplink maritime communication framework in which a UAV equipped with an MEC server cooperates with an onshore base station to serve a maritime user with stringent latency requirements.
- Formulate a joint optimization problem to minimize the total system delay by simultaneously determining the optimal flying altitude of the UAV and the optimal task allocation ratio between the UAV and the base station.
- Derive a semi closed-form expression for the optimal flying altitude and a closed-form expression for the optimal task allocation ratio to enable efficient and analytically tractable solutions.

- Demonstrate the impact of system parameters including transmit power, bandwidth, CPU frequency, and horizontal distance on the optimal altitude and task allocation ratio through simulation.

Methodology:

1. System Model:

A UAV-assisted MEC uplink maritime communication system is considered comprising a UAV, a maritime user, and an onshore base station, all equipped with a single antenna. The user simultaneously offloads a fraction α of the total computing task L to the UAV via the Sea-to-Air (S2A) link and the remaining fraction $1 - \alpha$ to the base station via the Sea-to-Ground (S2G) link using a Frequency Division Multiple Access (FDMA) scheme. The S2A channel adopts a probabilistic LoS/NLoS model in which the LoS probability is given by:

$$p_{\text{LoS}} = \frac{1}{1 + a \exp[-b(\theta - a)]},$$

where a and b are environment-dependent constants and $\theta = \arctan(H/r)$ is the elevation angle. The average path loss of the S2A channel is expressed as $\overline{PL} = p_{\text{LoS}}PL_{\text{LoS}} + p_{\text{NLoS}}PL_{\text{NLoS}}$. The S2G channel is modeled as Rayleigh fading with channel gain $h_{\text{S2G}} = \beta_0(d_{\text{S2G}})^{-\varphi}\xi_{\text{S2G}}$.

2. Delay Model and Problem Formulation:

The total system delay is modeled as $T = \max\{t_{\text{S2A}} + t_{\text{UAV}}, t_{\text{S2G}} + t_{\text{BS}}\}$, where the transmission delays are $t_{\text{S2A}} = \alpha L/R_{\text{UAV}}$ and $t_{\text{S2G}} = (1 - \alpha)L/R_{\text{BS}}$, and the computation delays are $t_{\text{UAV}} = \alpha LC/f_{\text{UAV}}$ and $t_{\text{BS}} = (1 - \alpha)LC/f_{\text{BS}}$, with C denoting the number of CPU cycles per bit. The joint optimization problem is formulated as:

$$\min_{\{H, \alpha\}} T \quad \text{subject to} \quad 0 \leq \alpha \leq 1, \quad H_{\min} \leq H \leq H_{\max}.$$

3. Optimal Altitude Derivation:

The optimal flying altitude H^{opt} is obtained by minimizing the S2A path loss function $g(H) = (H^2 + r^2) \left[\eta_{\text{NLoS}} + \frac{\eta_{\text{LoS}} - \eta_{\text{NLoS}}}{1 + at(H)} \right]$, where $t(H) = \exp \left[-b \left(\frac{180}{\pi} \arctan \left(\frac{H}{r} \right) - a \right) \right]$. Setting the first-order derivative $g'(H) = 0$ yields the semi closed-form expression:

$$H^{\text{opt}} = \frac{-At(H)}{2\eta_{\text{NLoS}}[1 + at(H)]^2 + 2(\eta_{\text{LoS}} - \eta_{\text{NLoS}})[1 + at(H)]},$$

which is solved iteratively and depends only on the horizontal distance r between the UAV's ground projection and the user.

4. Optimal Task Allocation Derivation:

Since $T_1 = \alpha \left(\frac{L}{R_{\text{UAV}}} + \frac{LC}{f_{\text{UAV}}} \right)$ is monotonically increasing in α and $T_2 = (1-\alpha) \left(\frac{L}{R_{\text{BS}}} + \frac{LC}{f_{\text{BS}}} \right)$ is monotonically decreasing in α , the total delay T is minimized when $T_1 = T_2$, yielding the closed-form optimal task allocation ratio:

$$\alpha^{\text{opt}} = \frac{\frac{1}{R_{\text{BS}}} + \frac{C}{f_{\text{BS}}}}{\frac{1}{R_{\text{UAV}}} + \frac{C}{f_{\text{UAV}}} + \frac{1}{R_{\text{BS}}} + \frac{C}{f_{\text{BS}}}} .$$

5. Simulation and Validation:

Numerical simulations are conducted with parameters including horizontal distance $r = 50$ m, carrier frequency $f_c = 2.4$ GHz, bandwidth $B_{\text{UAV}} = B_{\text{BS}} = 1$ MHz, and CPU frequencies $f_{\text{UAV}} = 1$ GHz and $f_{\text{BS}} = 10$ GHz. The analytically derived optimal values $H^{\text{opt}} = 57$ m and $\alpha^{\text{opt}} = 0.62$ are verified against one-dimensional exhaustive search results. The impact of transmit power, bandwidth, CPU frequency, and distance parameters on the optimal altitude and task allocation ratio are systematically analyzed.

Key Results:

- The optimal flying altitude H^{opt} is solely determined by the horizontal distance r between the UAV and the user, and is independent of transmit power, bandwidth, and CPU frequency.
- The optimal task allocation ratio α^{opt} increases with higher UAV bandwidth B_{UAV} or UAV CPU frequency f_{UAV} , and decreases with higher base station bandwidth B_{BS} or base station CPU frequency f_{BS} .
- Analytical results derived from the semi closed-form and closed-form expressions closely match the one-dimensional exhaustive search results, confirming the accuracy of the theoretical analysis.
- The proposed cooperative UAV-BS offloading framework effectively minimizes total system latency compared to single-node offloading approaches.

Limitations:

- The system model considers only a single maritime user; extension to multi-user scenarios with interference management and dynamic task scheduling remains unaddressed.
- The UAV is assumed to be stationary at a fixed horizontal position during communication, and trajectory optimization for a mobile UAV serving multiple users is outside the scope of this work.

- The probabilistic LoS channel model uses environment-specific constants a and b calibrated for suburban or open-sea environments, which may not accurately represent dense urban or mountainous terrain UAV deployments.
- The framework does not address UAV energy consumption or battery constraints, which are critical for determining feasible operational altitudes and flight durations in practical maritime deployments.
- The height-based offloading concept in this paper considers a static single-UAV scenario, whereas the present project extends this to a dynamic multi-UAV framework operating across varying altitudes from 50–200 m with adaptive base station load balancing.

2.1.8 Optimal LAP Altitude for Maximum Coverage [8]

Problem Addressed:

- Fixed cellular network infrastructure is highly vulnerable to natural disasters such as floods, earthquakes, and tsunamis, creating an urgent need for rapid, cost-effective temporary wireless recovery solutions in affected areas.
- Low Altitude Platforms (LAPs) are identified as promising airborne base stations for emergency coverage deployment; however, the number of deployable LAPs is severely limited during disaster aftermath, mandating full exploitation of each deployed platform through altitude optimization.
- Existing Air-to-Ground (ATG) propagation models do not provide a tractable closed-form expression for the LoS probability between a LAP and a ground receiver as a function of elevation angle and urban environment parameters, limiting the ability to analytically optimize LAP altitude.
- No analytical model existed prior to this work for determining the optimal LAP altitude that maximizes ground coverage radius, given the maximum allowable pathloss threshold and the statistical characteristics of the urban deployment environment.

Objective:

- Develop a mathematical model for predicting the probability of geometrical Line-of-Sight (LoS) between a LAP and a ground receiver as a closed-form function of the elevation angle and ITU-defined urban environment statistical parameters.

- Derive an analytical approach to determine the optimal LAP altitude h_{OPT} that maximizes the ground coverage radius R for a given maximum allowable pathloss PL_{max} .
- Generalize the LoS probability model across four representative urban environments Sub-urban, Urban, Dense Urban, and Highrise Urban by linking the S-curve fitting parameters directly to environment variables through surface polynomial fitting.
- Identify the optimal elevation angle θ_{OPT} as an environment-dependent invariant that maintains a constant ratio h_{OPT}/R regardless of the maximum allowed pathloss.

Methodology:

1. ATG Propagation Model:

Radio signals from a LAP base station propagate through free space before incurring shadowing and scattering from urban structures. The mean ATG pathloss for each propagation group $\xi \in \{\text{LoS}, \text{NLoS}\}$ is modeled as:

$$\text{PL}_{\xi} = \text{FSPL} + \eta_{\xi},$$

where FSPL is the free-space pathloss and η_{ξ} is the mean excessive pathloss of propagation group ξ . The spatial expectation of pathloss Λ between a LAP and all ground receivers at a common elevation angle θ is:

$$\Lambda = \sum_{\xi} \text{PL}_{\xi} \cdot \mathbf{P}(\xi, \theta),$$

with $\mathbf{P}(\text{NLoS}, \theta) = 1 - \mathbf{P}(\text{LoS}, \theta)$.

2. Closed-Form LoS Probability Model:

Following the ITU recommendation, the geometrical LoS probability between a LAP and a ground receiver in an urban environment is characterized by three statistical parameters: α (ratio of built-up area to total land area), β (mean number of buildings per unit area in buildings/km²), and γ (scale parameter of the Rayleigh distribution of building heights). The resulting LoS probability series from the ITU formulation is approximated by a modified Sigmoid S-curve:

$$\mathbf{P}(\text{LoS}, \theta) = \frac{1}{1 + a \exp(-b[\theta - a])},$$

where a and b are S-curve parameters linked to the environment variables α , β , and γ through a two-variable surface polynomial fitting of the form:

$$z = \sum_{j=0}^3 \sum_{i=0}^{3-j} C_{ij} (\alpha\beta)^i \gamma^j,$$

with polynomial coefficients C_{ij} tabulated for both parameters a and b .

3. Optimal Altitude Derivation:

The coverage radius R is defined as the ground radius within which the mean pathloss Λ does not exceed $\text{PL}_{\max}$. Substituting the LoS probability model and the pathloss expressions into the coverage equation yields the implicit relationship:

$$\text{PL}_{\max} = \frac{A}{1 + a \exp(-b[\arctan(\frac{h}{R}) - a])} + 10 \log(h^2 + R^2) + B,$$

where $A = \eta_{\text{LoS}} - \eta_{\text{NLoS}}$ and $B = 20 \log f + 20 \log(4\pi/c) + \eta_{\text{NLoS}}$. The optimal LAP altitude h_{OPT} is found by numerically solving the critical point condition $\partial R / \partial h = 0$.

4. Optimal Elevation Angle:

Rewriting the coverage equation in terms of elevation angle θ and radius R , the optimum elevation angle θ_{OPT} is obtained by solving $\partial R / \partial \theta = 0$:

$$\frac{\pi}{9 \ln(10)} \tan(\theta_{\text{OPT}}) + \frac{abA \exp(-b[\theta_{\text{OPT}} - a])}{[a \exp(-b[\theta_{\text{OPT}} - a]) + 1]^2} = 0.$$

This solution is independent of $\text{PL}_{\max}$ and unique for a given set of environment parameters (a, b, A) , confirming that a constant ratio h_{OPT}/R holds across all pathloss thresholds.

5. Simulation and Validation:

The model is evaluated for four urban environments Suburban (0.1, 750, 8), Urban (0.3, 500, 15), Dense Urban (0.5, 300, 20), and Highrise Urban (0.5, 300, 50) for (α, β, γ) respectively. Simulations use $\text{PL}_{\max} = 10$ dB and $f = 2000$ MHz. The cell radius versus LAP altitude curves are plotted for each environment and the optimal altitudes are determined numerically from the critical point condition.

Key Results:

- The optimal LAP altitude h_{OPT} is strongly dependent on the urban environment category; denser environments with taller buildings require higher optimal altitudes to achieve maximum LoS coverage.
- The LoS probability Sigmoid approximation closely fits the ITU-derived series for all four urban environments, enabling tractable analytical optimization.
- A constant optimum elevation angle θ_{OPT} exists for each environment that maintains a fixed ratio h_{OPT}/R , independent of the maximum allowed pathloss threshold.

- The model provides a computationally efficient closed-form tool for rapid LAP deployment planning in emergency and disaster recovery scenarios without requiring exhaustive numerical search.

Limitations:

- The model assumes a quasi-stationary LAP hovering at a fixed altitude, and does not address trajectory optimization for a mobile UAV serving users distributed across a large geographic area.
- Small-scale fading effects and large-scale channel variations are not incorporated; the model operates solely on mean pathloss, which may underestimate instantaneous link degradation in practice.
- The coverage analysis considers a single LAP serving an omnidirectional coverage disk; multi-UAV cooperative deployment with overlapping coverage zones, inter-cell interference, and load balancing is outside the scope of this work[12].
- The S-curve approximation parameters a and b are calibrated for ITU-defined urban environments and may not accurately represent non-standard or mixed terrain deployments encountered in the present project's $2000 \text{ m} \times 2000 \text{ m}$ simulation area with three ground base stations.
- The present project extends the LoS probability model from this work to govern altitude-dependent offloading decisions in a dynamic multi-UAV handover framework, addressing the gap between static single-LAP coverage optimization and real-time cooperative aerial network management.

2.2 Summary and Gaps Identified

Table 2.1: Comparative Summary of Literature Survey

Paper	Objective & Method	Advantages	Disadvantages
Li et al., 2021 – <i>“The Evolution of Networks and Management in a 6G World”</i>	Discusses fast and reliable handovers in 6G and UAV-enabled networks using conceptual analysis of next-generation network management and mobility mechanisms.	Highlights seamless connectivity importance. Provides insights into mobility management challenges in 6G.	Mostly conceptual. No detailed implementation strategies for UAV handovers.
Gupta et al., 2016 – <i>“Survey of Important Issues in UAV Communication Networks”</i>	Comprehensive survey of UAV communication architectures and networking challenges. Discusses mobility, connectivity constraints, and UAV integration in wireless systems.	Broad understanding of UAV communication networks. Identifies major challenges affecting UAV connectivity.	Does not consider movement direction or trajectory parameters for handover decisions.
Taleb et al., 2019 – <i>“Towards Smart 6G: Mobility Prediction for Dynamic Edge Services Migration”</i>	ML-based mobility prediction considering direction and trajectory of moving nodes, addressing limitations of traditional 2D handover models.	Improves mobility awareness and prediction accuracy. Enables better service migration decisions in dynamic environments.	Does not fully consider UAV velocity variations affecting handover frequency.
Zhang et al., 2021 – <i>“Proactive Handover Decision for UAVs with Deep Reinforcement Learning”</i>	Applies Deep Reinforcement Learning using Proximal Policy Optimization (PPO). Proactively manages handovers for high-speed UAVs.	Improves mobility awareness and prediction accuracy. Better service migration in dynamic environments.	Does not fully consider UAV velocity variations affecting handover frequency.
Błaszczyszyn et al., 2013 – <i>“Spatial Disparity of QoS Metrics Between Base Stations in Wireless Cellular Networks”</i>	Uses queueing theory to analyze QoS variation due to uneven base station loading.	Analyzes network performance analytically. Identifies QoS degradation caused by overloaded base stations.	Focuses on ground cellular networks only. No UAV mobility or intelligent handover mechanisms.
Wang et al., 2022 – <i>“Explainable AI for UAV Mobility Management: A Deep Q-Network Approach for Handover Minimization”</i>	Uses Deep Q-Network (DQN) for UAV handover decisions. Integrates Explainable AI (SHAP) to interpret decision making.	Reduces unnecessary handovers. Improves transparency of AI-based decisions.	Requires training data and high computational resources. Does not consider altitude as a deployment parameter.

Paper	Objective & Method	Advantages	Disadvantages
Chen et al., 2020 – “ <i>Altitude Optimization and Task Allocation of UAV-Assisted MEC Communication System</i> ”	Develops a mathematical optimization model for optimal UAV relay altitude selection. Allocates computing tasks between UAV and ground stations in MEC systems.	Reduces communication delay. Enables UAVs to function as mobile edge computing nodes.	Designed mainly for maritime environments. Limited generalizability to other scenarios.
Al-Hourani et al., 2014 – “ <i>Optimal LAP Altitude for Maximum Coverage</i> ”	Derives a closed-form mathematical model to determine optimal altitude for low-altitude platforms (LAPs) based on urban environment parameters.	Provides a simple guideline for UAV deployment height. Maximizes ground coverage area.	Sensitive to environmental factors such as buildings and signal attenuation. Does not consider mobility-aware handover mechanisms.

Research Gaps Identified

The reviewed literature collectively establishes the importance of adaptive handover, mobility-aware communication, and altitude optimization in UAV networks. The following gaps motivate the present work:

1. RSSI-only handover methods lack adaptivity to 3D UAV mobility patterns. Conventional threshold-based handover schemes that rely solely on Received Signal Strength Indicator (RSSI) fail to account for the altitude-induced signal variations and rapid topology changes inherent to 3D UAV flight, resulting in frequent ping-pong effects and unnecessary handovers.
2. ML and RL approaches incur significant computational overhead that limits deployment on resource-constrained UAV platforms. While Deep Reinforcement Learning and LSTM-based prediction frameworks demonstrate strong performance in simulation, their training complexity and inference latency render them impractical for real-time deployment on battery-powered UAVs with limited onboard processing capacity.
3. Altitude is rarely treated as an active variable in handover decision-making, despite its fundamental influence on Line-of-Sight (LoS) probability. Existing handover frameworks predominantly operate in 2D horizontal space and do not leverage the altitude-dependent Air-to-Ground channel characteristics modeled by the probabilistic LoS framework of Al-Hourani et al, leaving a critical dimension of UAV communication unexploited.
4. Cooperative offloading between drones operating at different altitudes has not been sys-

tematically integrated with the handover framework. Although UAV-assisted MEC offloading has been studied independently, no existing work combines altitude-aware relay selection with a real-time handover engine to provide seamless fallback connectivity when primary base station association fails.

5. A unified priority-driven decision engine combining load balancing, signal quality, and altitude-aware relay selection has not been explored. The surveyed literature addresses these dimensions in isolation; no framework simultaneously enforces tiered handover priorities urgent, signal-based, and load-based while dynamically adjusting base station association thresholds based on UAV altitude and cooperative offloading availability.

These identified gaps directly motivate the design of the proposed adaptive multi-criteria handover and cooperative offloading framework presented in this work, which addresses each limitation through a lightweight, simulation-validated architecture operating across a population of 5 to 50 UAVs in a $2000 \text{ m} \times 2000 \text{ m}$ environment with three ground base stations.

2.3 Summary of Chapter

This chapter reviewed eight key papers and identified five specific gaps particularly the absence of a priority-driven multi-criteria handover engine integrated with altitude-aware cooperative offloading that directly motivate the system design presented in the following chapter.

Chapter 3

System Design

3.1 Architecture Overview

The proposed system architecture for enabling handover and energy efficiency in UAV networks adopts a two-layered air-ground framework designed to ensure real-time connectivity, priority-driven decision-making, and cooperative relay offloading. The system integrates a multi-criteria decision engine with altitude-aware relay selection and adaptive base station association, operating across an Air Layer and a Ground Layer within a $2000 \text{ m} \times 2000 \text{ m}$ simulation environment.

The architecture consists of the following layers:

- Air Layer (Drone UE and UAV-BS Relay Nodes)
- Ground Layer (Base Station Infrastructure)
- Decision Engine (Handover and Relay Control Logic)

3.1.1 Air Layer

The Air Layer comprises the primary Drone UE and a set of UAV-BS relay nodes distributed across the simulation area. Each Drone UE is characterized by its instantaneous RSSI value (RSSI_0) and its ground distance (dist_0) to the serving base station. The Drone UE maintains a serving link to its currently associated ground base station and is continuously monitored by the decision engine for handover eligibility.

Two UAV-BS relay nodes UAV-BS 1 and UAV-BS 2 operate within the air layer as cooperative offloading intermediaries. Each UAV-BS relay maintains a capacity parameter $C_{\text{UAV}} = 8$, defining the maximum number of Drone UEs it can simultaneously serve as a relay. UAV-BS 1 operates within a coverage range of $\leq 1000 \text{ m}$ from the Drone UE and accepts relay offload connections when ground base stations are unavailable or overloaded. A *pendingHO poll* of 0.1 s governs the relay association latency, ensuring near-instantaneous fallback connectivity. UAV-BS 2 serves as a secondary relay option when UAV-BS 1 has reached its capacity threshold

($C_{\text{UAV}} = 8$).

3.1.2 Ground Layer

3.1.3 Base Station States

Each base station in the Ground Layer dynamically transitions between three possible operational states — *Available*, *Serving*, and *Full* — based on the number of currently connected drones relative to the maximum capacity $C_{\text{BS}} = 10$. The state of each base station is continuously monitored and updated by the handover decision engine at every position update interval of 0.1 s.

State 1: Available (Empty)

A base station is in the *Available* state when it has no connected drones or has remaining capacity to accept new connections. This is the default initial state of all base stations at simulation start.

Condition:

$$0 \leq \text{Load}(\text{BS}) < C_{\text{BS}}$$

Characteristics:

The base station is eligible to receive incoming drone connections or handover requests.

The BS Search module will consider this station as a valid handover target candidate.

The station is ranked by load during candidate selection: $\text{BS}_{\text{cand}} = \arg \min_{b \in \text{BS}_{\text{set}}} \text{Load}(b)$

A base station with zero connected drones ($\text{Load}(\text{BS}) = 0$) is considered *empty* and will always be prioritized as the first handover target.

Signal strength from the BS must still satisfy: $\text{RSSI}(\text{BS}) \geq -78 \text{ dBm}$ for the drone to associate with it.

Decision Engine Outcome: `H0_SUCCESS` when selected as target.

State 2: Serving (Partially Loaded)

A base station is in the *Serving* state when it has one or more connected drones but has not yet reached its maximum capacity. This is the normal operational state during active simulation.

Condition:

$$0 < \text{Load}(\text{BS}) < C_{\text{BS}}$$

Characteristics:

The base station is actively serving connected drones while remaining eligible to accept additional connections.

Drones currently associated with this BS maintain their link as long as:

$$\text{dist}_0 \leq \text{range}(\text{BS}_{\text{cur}})$$

$$\text{RSSI}_0 \geq -95 \text{ dBm}$$

No handover trigger (urgent, signal-based, or load-based) is active.

A load-based handover trigger fires when the load difference between this BS and a candidate BS satisfies: $\Delta\text{Load} \geq \Delta L = 3$, redistributing drones to balance the network.

The BS remains a valid handover target as long as $\text{Load}(\text{BS}) < C_{\text{BS}}$.

Decision Engine Outcome: NO_HO_NEEDED for currently served drones; HO_SUCCESS when selected as target for incoming drones.

State 3: Full (At Capacity)

A base station is in the *Full* state when the number of connected drones has reached the maximum capacity $C_{\text{BS}} = 12$. No additional connections can be accepted.

Condition:

$$\text{Load}(\text{BS}) = C_{\text{BS}} = 10$$

Characteristics:

- The base station is excluded from the handover target candidate set by the BS Search module.
- Any drone attempting to hand over to a *Full* BS is redirected to Phase 4 (UAV-BS Relay Search) of the decision pipeline.
- If no relay node is available either, the drone is logged as BLOCKED and queued until a capacity slot is freed.
- The BS exits the *Full* state and transitions back to *Serving* when a connected drone disconnects or hands over to another station, decrementing the load: $\text{Load}(\text{BS}) \leftarrow \text{Load}(\text{BS}) - 1$

Decision Engine Outcome: BLOCKED for incoming handover requests; existing connections are maintained.

State Transition Table

Table 3.1 summarizes all three base station states, their load conditions, eligibility for handover, and the corresponding decision engine outcomes.

Table 3.1: Base Station State Transition Summary

State	Load Condition	Accept New	HO Target	Outcome
<i>Available</i> (Empty)	$0 \leq \text{Load} < C_{\text{BS}}$	Yes	Eligible	HO_SUCCESS
<i>Serving</i> (Partial)	$0 < \text{Load} < C_{\text{BS}}$	Yes	Eligible	HO_SUCCESS NO_HO_NEEDED
<i>Full</i> (At Capacity)	$\text{Load} = C_{\text{BS}} = 12$	× No	× Excluded	BLOCKED → Relay Search

State Transition Diagram

The base station transitions between states as drones connect and disconnect throughout the simulation, as described below:

- **Available** → **Serving**: A drone successfully associates with or hands over to this BS, incrementing the load: $\text{Load} \leftarrow \text{Load} + 1$
- **Serving** → **Full**: The load reaches maximum capacity as additional drones connect: $\text{Load} = C_{\text{BS}} = 12$
- **Full** → **Serving**: A connected drone disconnects or hands over to another BS, freeing a capacity slot: $\text{Load} \leftarrow \text{Load} - 1$

- **Serving $\rightarrow$ Available:** All connected drones disconnect or hand over, leaving the BS with no active connections: Load = 0

The Ground Layer consists of three base stations providing primary wireless coverage to Drone UEs operating within the simulation area:

- **BS₁ (Serving):** The currently associated base station of the Drone UE, operating with a capacity of $C_{BS} = 12$. The Drone UE maintains its serving link to BS₁ until a handover trigger is detected by the decision engine.
- **BS₂ (Target):** A candidate target base station identified by the BS Search module. BS₂ is selected when its load satisfies the condition $\text{load} < C_{BS}$, ensuring sufficient capacity to accommodate the incoming Drone UE. A successful handover to BS₂ is indicated by the *HO_SUCCESS* event.
- **BS₃ (Full):** A base station operating at maximum capacity, with $\text{load} = C_{BS}$. BS₃ is excluded from target selection by the BS Search module, and any handover attempt directed toward it results in a *BLOCKED* outcome.

The RSSI thresholds governing ground layer association are defined as -78 dBm (strong signal), -93.5 dBm (marginal signal), and -95 dBm (minimum acceptable threshold), as indicated on the signal quality scale in the system diagram.

3.1.4 Decision Engine

The Decision Engine forms the core control logic of the proposed framework and operates through four sequential processing stages:

1. **Connectivity Validation:** The first stage evaluates whether the Drone UE has exceeded the maximum permissible range ($\text{dist} > \text{range}$) or has fallen below the minimum RSSI threshold ($\text{RSSI} < -95$ dBm). If either condition is satisfied, the Drone UE is immediately issued a *DISCONNECT* command, indicating an out-of-range or critically degraded link condition.
2. **Trigger Evaluation:** The second stage classifies the handover trigger into one of three priority categories **Urgent**, **Load**, or **Signal** based on the instantaneous RSSI value and its rate of change (ΔL). The evaluation thresholds are set at -93.5 dBm for marginal signal quality and $\Delta L \geq 3$ for load-based triggers, with -78 dBm defining the upper

boundary for signal-based handover initiation. If the trigger is classified as a relay offload condition, the Drone UE is queued on the UAV-BS relay as a *RELAY_OFFLOAD* outcome.

3. **BS Search:** The third stage performs a target base station search using the criterion $\text{argmin load}(b)$ subject to $\text{load} < C_{\text{BS}} = 12$. The base station with the minimum current load that satisfies the capacity constraint is selected as the handover target, resulting in a *HO_SUCCESS* event upon successful association.
4. **Relay Search (Fallback):** If no eligible ground base station is identified during the BS Search stage, the decision engine initiates a fallback Relay Search. This stage identifies available UAV-BS relay nodes within a distance of $d \leq 1000$ m with $\text{load} < C_{\text{UAV}}$, polling at 0.1 s intervals to establish a relay link to the ground BS. If neither a ground BS nor a UAV-BS relay is available, the Drone UE receives a *BLOCKED* status, indicating that no BS or UAV-BS is currently free.

3.1.5 System Workflow

The complete operational flow of the proposed system is as follows:

1. **Link Monitoring:** The Drone UE continuously transmits its RSSI and distance metrics to the Decision Engine via the serving link.
2. **Trigger Classification:** The Trigger Evaluation stage classifies the handover condition as Urgent, Load-based, Signal-based, or Relay Offload.
3. **Target Selection:** The BS Search stage identifies the least-loaded eligible ground base station as the handover target.
4. **Handover Execution:** Upon target confirmation, the handover is executed and a *HO_SUCCESS* event is recorded.
5. **Relay Fallback:** If no ground BS is available, the Relay Search stage identifies the nearest available UAV-BS relay node for cooperative offloading.
6. **Blocking:** If no ground BS or relay node can accommodate the Drone UE, a *BLOCKED* status is issued and the drone remains in a pending state until a resource becomes available.

3.1.6 Key Architectural Features

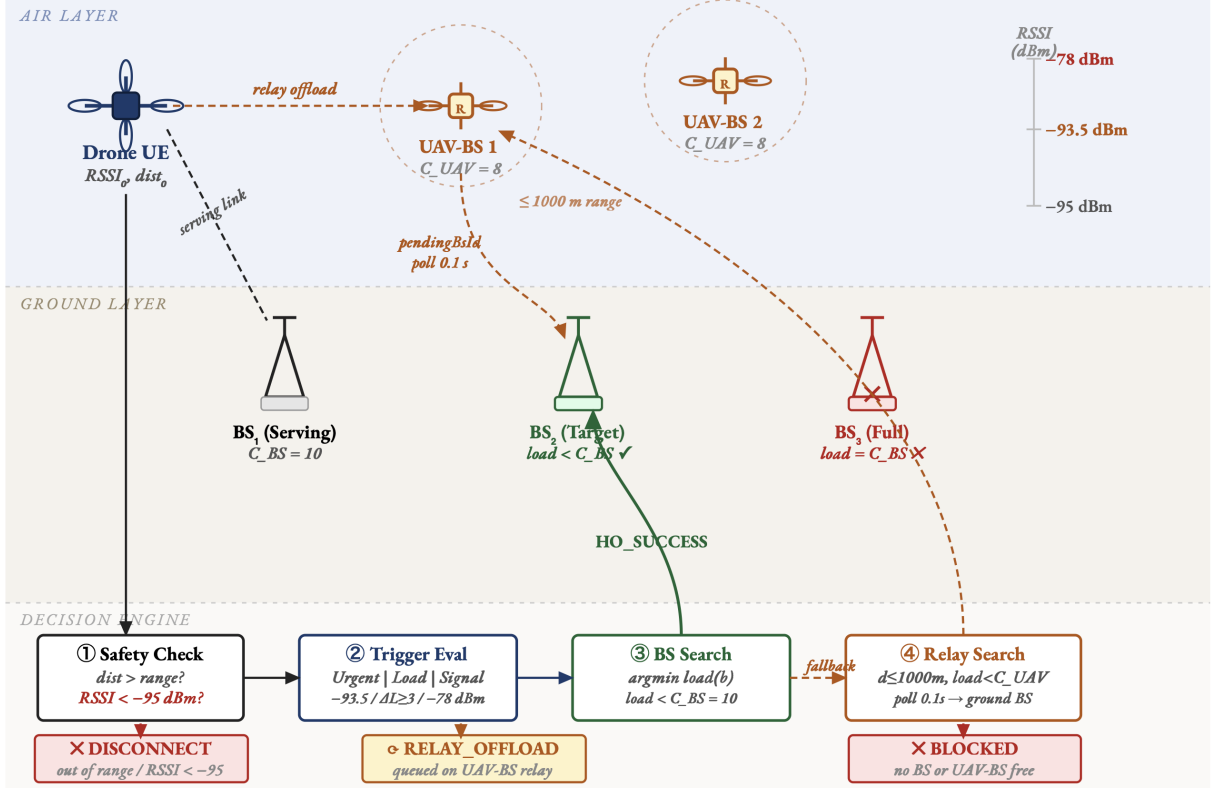


Figure 3.1: Proposed System Architecture for UAV Handover and Energy Efficiency

- Priority-driven four-stage decision engine combining safety, signal quality, load balancing, and cooperative relay fallback into a unified handover control framework.
- Altitude-aware cooperative offloading through UAV-BS relay nodes providing seamless fallback connectivity when ground base station capacity is exhausted.
- Capacity-constrained base station selection using a minimum-load criterion to prevent network congestion and ensure balanced resource utilization across all three ground base stations.
- Near-instantaneous relay association with a 0.1 s poll interval minimizing handover delay during fallback relay search operations.
- Dual-threshold RSSI monitoring at -93.5 dBm and -95 dBm enabling proactive handover initiation before critical link degradation occurs.

3.2 Component Design

Data Flow Diagram

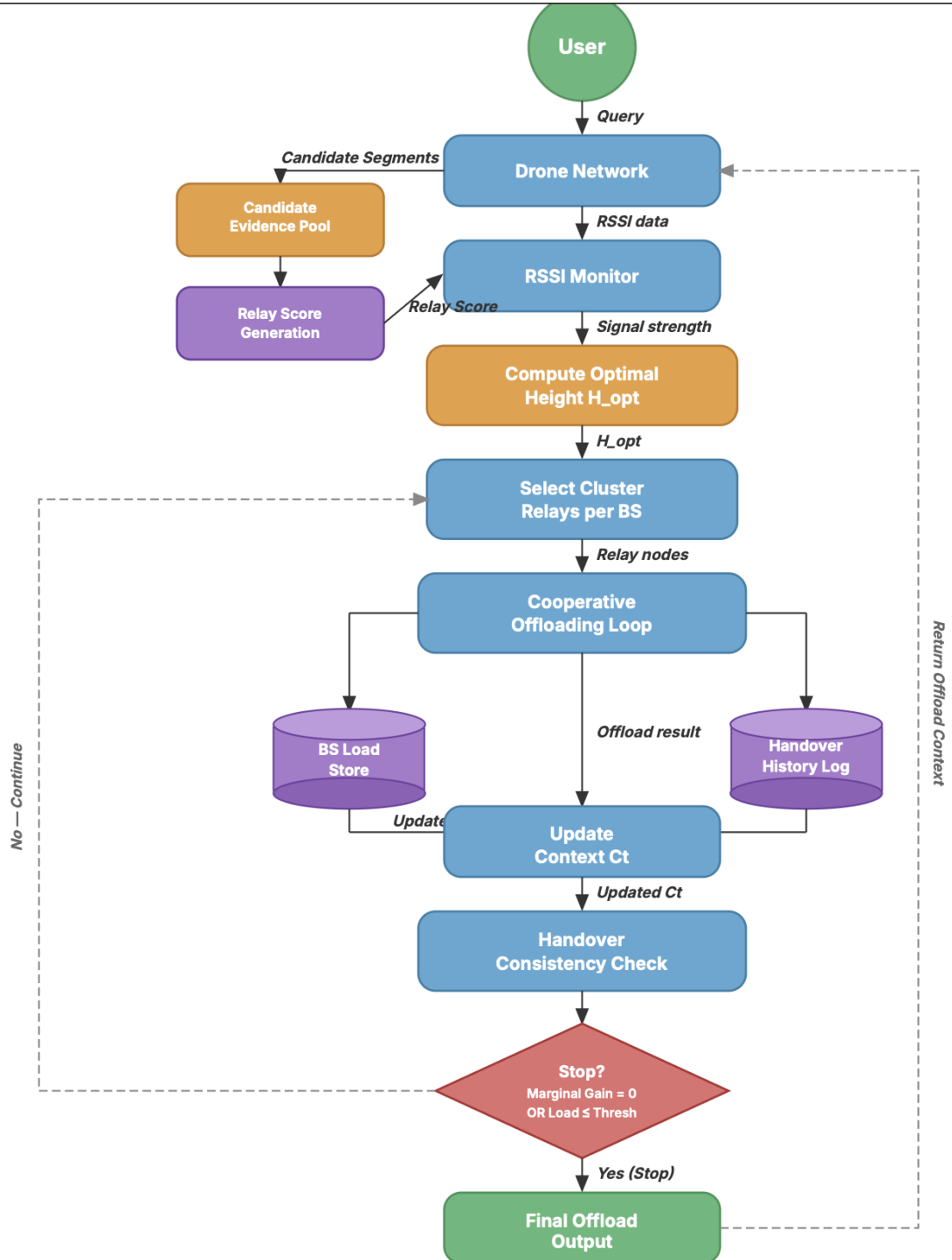


Figure 3.2: Dataflow Diagram

Figure 3.2 presents the data flow of the cooperative offloading scheme. The process begins with drones transmitting data to the nearest base station. The RSSI Monitor continuously evaluates signal quality, and drones falling below the offloading threshold (-80 dBm) are identified as weak-signal candidates.

For each such drone, an optimal relay altitude H_{opt} is computed using the ATG path-loss model, and the best-positioned relay drone within the BS cluster is selected based on a combined RSSI and altitude score. Weak-signal drones are then routed through the relay to the BS via the Cooperative Offloading Loop.

After each iteration, the BS load and handover history are updated. The loop continues until either no further throughput gain is achieved or the load is balanced across all base stations, at which point the Final Offload Output is returned to the network.

3.3 Sequence Diagram

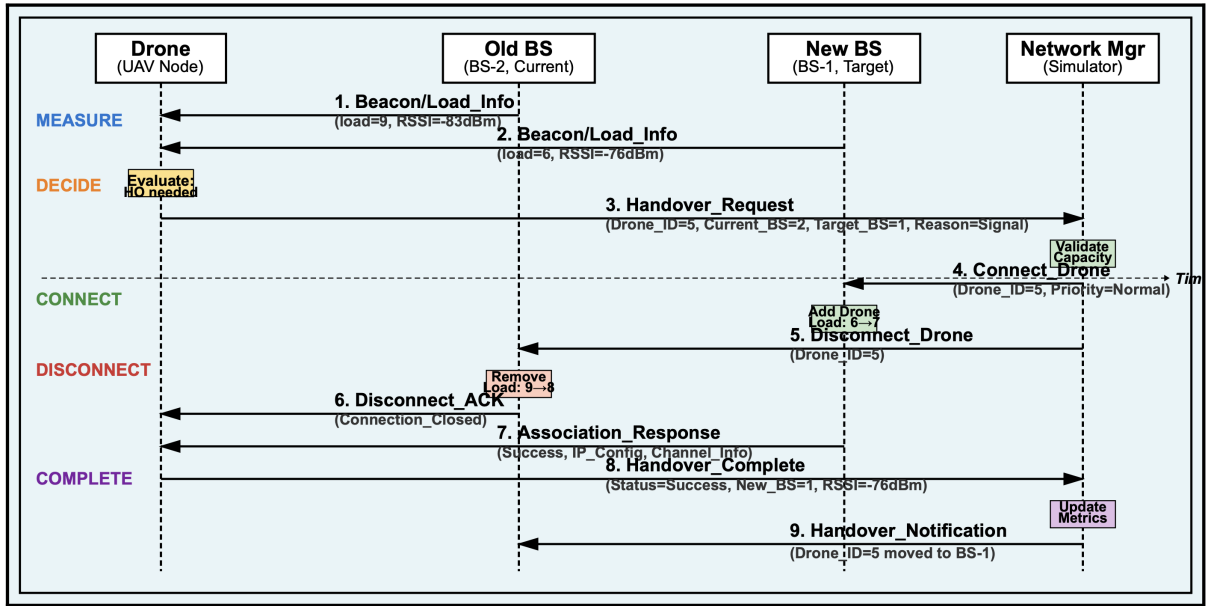


Figure 3.3: Sequence Diagram

The diagram involves four entities: the **Drone** (UAV Node), the **Old BS** (BS-2, Current Serving Base Station), the **New BS** (BS-1, Target Base Station), and the **Network Manager** (Simulator). The handover process unfolds across five distinct phases: Measure, Decide, Connect, Disconnect, and Complete.

3.3.1 Phase 1: Measure

The process begins with the drone receiving beacon and load information from both the currently serving base station and neighboring base stations. In step 1, the Old BS (BS-2) transmits a **Beacon/Load_Info** message to the drone, reporting a load of 9 and an RSSI of -83 dBm. In step 2, the New BS (BS-1) similarly broadcasts its **Beacon/Load_Info**, reporting a lower load of 6 and a stronger RSSI of -76 dBm. The drone collects and compares these measurements to assess the current connectivity conditions.

3.3.2 Phase 2: Decide

Based on the measured beacon information, the drone's onboard decision engine evaluates whether a handover is necessary. The drone identifies that BS-1 offers a superior RSSI (-76 dBm versus -83 dBm) and a lower load (6 versus 9), satisfying both the signal-based and load-based handover trigger conditions. The decision block is marked *Evaluate: HO needed*. In step 3, the drone dispatches a **Handover_Request** message to the Network Manager carrying the parameters: **Drone_ID=5**, **Current_BS=2**, **Target_BS=1**, and **Reason=Signal**. The Network Manager then validates the capacity of the target BS before proceeding.

3.3.3 Phase 3: Connect

Upon successful capacity validation, the Network Manager issues a **Connect_Drone** command (step 4) to the New BS (BS-1) with parameters **Drone_ID=5** and **Priority=Normal**. The New BS accepts the connection, incrementing its load from 6 to 7, as indicated by the *Add Drone Load: 6 $\rightarrow$ 7* annotation. In step 5, the New BS sends a **Disconnect_Drone** signal to the Old BS (BS-2), instructing it to release the drone's existing connection.

3.3.4 Phase 4: Disconnect

Upon receiving the disconnect instruction from the **Network Manager**, the Old BS (BS-2) releases the drone's connection and decrements its load from 9 to 8. In **Step 6**, BS-2 sends a **Disconnect_ACK** message to the Drone with status **Connection_Closed**, formally terminating the existing association between Drone ID=5 and BS-2.

In **Step 7**, the New BS (BS-1) sends an **Association_Response** to the Drone confirming successful connection, along with IP configuration and channel information (**Success**, **IP_Config**, **Channel_Info**), establishing BS-1 as the new serving base station for Drone ID=5.

3.3.5 Phase 5: Complete

With the new association established, the drone transmits a **Handover_Complete** message (step 8) to the Network Manager, reporting **Status=Success**, **New_BS=1**, and **RSSI=-76 dBm**. The Network Manager updates its internal metrics to reflect the new drone-BS association. Finally, in step 9, the New BS issues a **Handover_Notification** back to the Network Manager confirming that **Drone_ID=5** has been successfully moved to BS-1, completing the entire handover procedure.

3.4 Module Division

3.4.1 Module 1: Drone and Base Station Environment

The simulation environment models a realistic UAV communication scenario within a $2000\text{ m} \times 2000\text{ m}$ operational area. Three ground base stations are deployed at fixed positions with defined coverage radii, enabling clear delineation of coverage zones and overlaps. The 30-drone fleet is initialized at random positions within the simulation area.

Position updates occur at 10 Hz, providing smooth trajectories that closely approximate real-world UAV flight patterns. The mobility model implements a modified random waypoint scheme: each drone selects a random destination, travels toward it at variable speed, and upon arrival selects a new destination creating continuous, unpredictable movement that stress-tests the handover mechanism.

Drone altitude varies dynamically between 50 m and 200 m during flight, with altitude changes coupled to horizontal movement to simulate realistic climbing and descent profiles. This 3D mobility model ensures evaluation under conditions representative of actual deployments. The environment module provides the decision engine with current drone position, altitude, RSSI to each base station, and dwell time forming the primary inputs to handover decision-making.

3.4.2 Module 2: Drone Mobility and Handover Decision Engine

The handover decision engine implements a four-phase, prioritized decision procedure evaluating multiple handover criteria in order of urgency. The algorithm parameters are defined as follows: RX sensitivity = -95 dBm, urgent RSSI threshold = -93.5 dBm, weak RSSI threshold = -78 dBm, hysteresis = 3 dB, minimum dwell time = 5 seconds, BS capacity = 12, UAV relay capacity = 8, and relay range = 1000 m.

Phase 1: Connectivity Validation

Phase 1 checks whether the drone's position is outside the serving base station's coverage range, or whether RSSI has fallen below -95 dBm. If either condition holds, the drone is immediately disconnected and `DISCONNECT` is returned. This phase ensures drones are not maintained in associations providing no viable communication link.

Phase 2: Trigger Evaluation

Three Boolean flags determine whether a handover is warranted:

- **Urgent** (`urgent`): True if $\text{RSSI} < -93.5$ dBm, indicating the link is approaching failure.
- **Load-based** (`loadHO`): True if dwell time ≥ 5 s AND the load difference between the current BS and the least-loaded BS is ≥ 3 .
- **Signal-based** (`sigHO`): True if $\text{RSSI} < -78$ dBm (or `urgent` is true) AND a target BS offers ≥ 3 dB RSSI gain.

If none of the three flags are set, `NO_HO_NEEDED` is returned. This gate prevents unnecessary processing and avoids disruptive handovers when the current connection is adequate.

Phase 3: Ground BS Candidate Search

If a handover is warranted, the algorithm finds the optimal ground base station target among all BSs providing RSSI above the sensitivity threshold. The BS with minimum current load is selected. If the candidate has available capacity ($\text{load} < 12$), the handover is executed and `HO_SUCCESS` is returned. If the candidate is at full capacity, execution falls through to Phase 4.

Phase 4: UAV-BS Relay Search

When no suitable ground BS is available, the algorithm searches for a UAV-BS relay node within 1000 m of the requesting drone with RSSI above the sensitivity threshold and available capacity ($\text{load} < 8$). The relay with minimum load is selected. If found, the drone is enqueued for relay offloading (`RELAY_OFFLOAD`), with the relay node polling ground BSs every 0.1 s until a slot opens. If no relay is available, the drone is marked `BLOCKED`.

3.4.3 Module 3: Height-Based Communication and Cooperative Offloading

The height-based communication module exploits altitude as a resource for improving Line-of-Sight (LoS) probability and communication reliability, grounded in the Air-to-Ground path loss model. As drones move through the simulation area, their altitude continuously varies, producing dynamically changing propagation conditions.

Drones at higher altitudes benefit from reduced obstruction probability and higher LoS likelihood. Drones at lower altitudes face greater obstruction, weaker signal strength, and more frequent relay eligibility. When a drone's RSSI falls below -80 dBm and no direct handover to a ground BS is possible, the module identifies the nearest higher-altitude drone within 1000 m with sufficient relay capacity.

The lower-altitude drone forwards its data to the relay node, which offloads the data to the ground base station using its superior LoS link. The optimal relay altitude is computed dynamically using the Air-to-Ground path loss formula, ensuring relay drones are positioned at heights that maximize LoS probability for both the drone-to-relay and relay-to-BS segments. Once data is forwarded, the lower-altitude drone does not participate further in that communication cycle, and all remaining communication occurs between the relay and the base station.

3.4.4 Module 4: GUI and Visualization

The visualization module provides a real-time graphical display of simulation state, including drone positions, base station coverage zones, active connection lines, and handover event logs. Each drone is rendered in the color of its associated base station, enabling instant visual recognition of association states and handover transitions.

All handover events, relay activations, disconnect events, and blocked states are logged systematically with timestamps and drone identifiers, forming the basis of performance analysis. Real-time metrics including per-BS load, average fleet RSSI, and cumulative handover counts are displayed continuously throughout the simulation.

3.5 Tools and Technologies

The simulation and development environment for the proposed system was configured using industry-standard tools and frameworks suited for UAV network emulation and real-time visualization. The following tools and technologies were employed in the implementation of the proposed system:

- **Simulation:** NS-3 network simulator for algorithm evaluation under realistic propagation conditions.
- **Programming Language:** C++ for algorithm development within the NS-3 framework.
- **Operating System:** macOS / Linux (Ubuntu 22.04 LTS).
- **Propagation Model:** Air-to-Ground path loss model based on Al-Hourani et al. [8] (2014).
- **Visualization:** Netanim module for real-time drone tracking and event logging.

3.6 Work breakdown

The work was structured into four key areas architecture design and visualization, environment modeling, signal evaluation, and system implementation and testing with each member taking ownership of a distinct module while contributing to the overall integration of the UAV handover and energy efficiency framework.

Architecture Design & Visualization

- Designed the overall system architecture of the UAV communication framework.
- Defined the interaction between drones, base stations, and the handover decision engine.
- Developed the association mechanism between drones and base stations.
- Implemented visualization of drone connectivity and handover events using the graphical interface.

Environment Modeling

- Developed the simulation environment representing the UAV communication network.
- Modeled drone mobility behavior and movement patterns within the simulation area.
- Configured base station coverage zones and network parameters.
- Created the environment required to evaluate connectivity and handover scenarios.

Signal Evaluation

- Implemented the signal evaluation mechanism for analyzing communication links.
- Monitored RSSI values and signal strength thresholds between drones and base stations.
- Evaluated signal conditions to determine when handover should be triggered.
- Supported the development of reliable signal-based handover decisions.

System Implementation & Testing

- Implemented the core system algorithms for handover decision-making.
- Integrated different modules including mobility, signal evaluation, and association logic.
- Performed system testing under different network conditions.
- Validated system performance and ensured stable connectivity during drone movement.

3.7 Key Deliverables

- Priority-based handover decision engine supporting urgent, signal-based, and load-based modes.
- Height-based cooperative offloading module with Air-to-Ground LoS modeling.
- Real-time GUI for drone movement visualization and handover event logging.
- Performance evaluation comparing load-based handover and cooperative offloading across 5–50 drone scenarios.

3.8 Project Timeline

The Gantt chart illustrates the project timeline for the UAV Handover project, spanning from August 2025 to March 2026, with tasks distributed across four development phases. The project commenced with a Literature Survey conducted collectively by all team members in August 2025, establishing the theoretical foundation for the work.

This was followed by Architecture Design in September 2025, defining the overall system structure and interaction between drones, base stations, and the handover decision engine. In

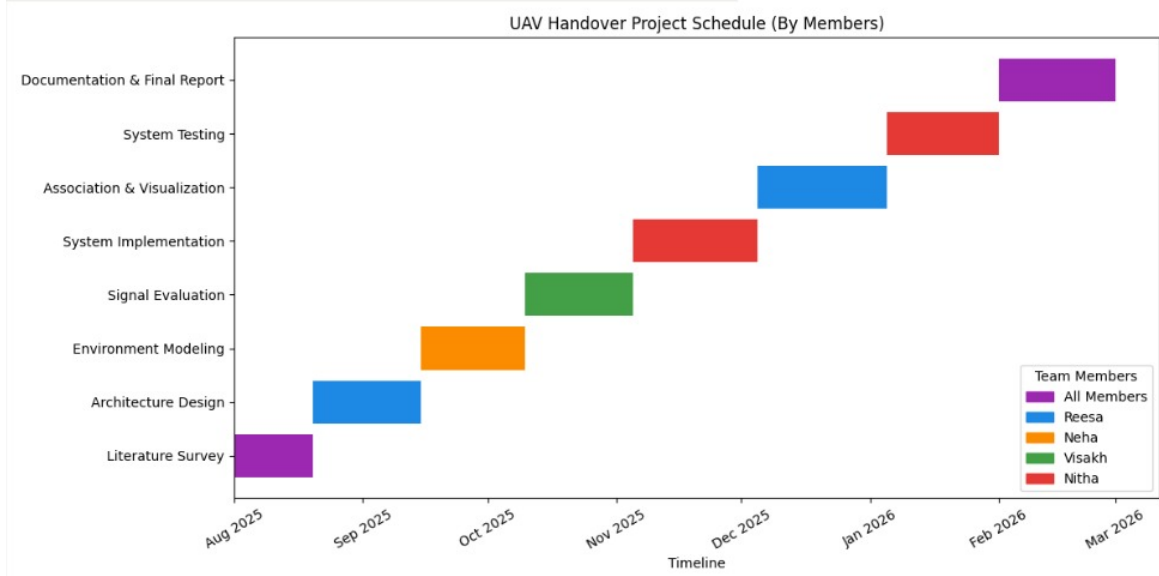


Figure 3.4: Gantt Chart

October 2025, the Environment Modeling phase was completed, developing the UAV simulation environment and configuring base station coverage zones. Signal Evaluation was then conducted during October–November 2025, implementing RSSI-based monitoring and signal threshold analysis to support handover decision-making.

System Implementation followed in November–December 2025, integrating the core handover algorithms and combining all developed modules into a unified system. The Association & Visualization phase was completed in December 2025–January 2026, developing the graphical interface for real-time drone connectivity and handover event visualization. System Testing was carried out in January–February 2026, validating performance under varying network conditions. Finally, Documentation & Final Report was completed collaboratively by all members from February to March 2026, consolidating the findings and results of the entire project.

3.9 Summary of chapter

This chapter described the layered system architecture, functional module decomposition, tools used, team work breakdown, and project schedule. The following chapter provides a detailed technical description of the implementation of each module.

Chapter 4

Proposed Method

This chapter presents the proposed methodology for enabling reliable handover and energy-efficient communication in UAV networks. The system considers a network of 30 drones operating dynamically across the coverage regions of 3 deployed ground base stations, with full 3D mobility support including altitude variation between 50 m and 200 m.

4.1 System Description

In the proposed system, 30 drones move within a $2000\text{ m} \times 2000\text{ m}$ network area where 3 base stations are deployed. As drones move, each UAV continuously stores and updates its position information, including its location and altitude. Using this position information, the system monitors how drones move across the different coverage regions of the base stations.

When a drone moves closer to another base station or its communication quality changes, the system evaluates whether a handover is required. Handover decisions are handled step-by-step using priority conditions, ensuring drones always remain connected to the most suitable base station. Since drones operate in a 3D environment, their altitude also affects communication quality.

The system therefore calculates an optimal altitude using the Air-to-Ground (A2G) path loss concept to improve Line-of-Sight (LoS) communication. When a drone flying at a lower altitude experiences weak connectivity, it forwards its data to a drone at a higher altitude. The higher-altitude drone, which usually has better LoS with the base station, then offloads the data to the base station. In this way, the system maintains stable communication, reduces connection loss, and ensures efficient data transmission while drones move dynamically.

4.2 Priority-Based Handover Decision Algorithm

The handover decision engine operates as a four-phase priority pipeline. At each update interval, the system evaluates the drone's current signal strength, distance, and network conditions to

determine the appropriate action. The algorithm inputs and outputs are defined as follows:

Input: UE, RSSI₀, dist₀, *BS_set*, *UAV_set*

Output: HO_SUCCESS | RELAY_OFFLOAD | BLOCKED | DISCONNECT

The system parameters governing the decision logic are listed in Table 4.1.

The four phases of the algorithm are described as follows:

1. **Phase 1 - Connectivity Validation:** The system first checks whether the drone has moved out of range of its serving base station or whether its signal strength has fallen below the receiver sensitivity limit of -95 dBm. If either condition holds, the drone is immediately disconnected.

Foundation: RSSI Computation (Used Across All Phases)

- **3D Distance**

$$d_{3D} = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

- **Horizontal Distance**

$$d_h = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

- **Elevation Angle**

$$\theta = \tan^{-1} \left(\frac{|\Delta z|}{d_h} \right) \times \frac{180}{\pi}$$

- **LoS Probability**

$$P_{\text{LoS}} = \frac{1}{1 + ae^{-b(\theta-a)}}$$

- **Free Space Path Loss**

$$\text{FSPL} = 20 \log_{10}(d_{3D}) + 20 \log_{10}(f) - 27.55$$

- **ATG Mean Path Loss**

$$\text{PL}_{\text{ATG}} = \text{FSPL} + P_{\text{LoS}}\eta_{\text{LoS}} + (1 - P_{\text{LoS}})\eta_{\text{NLoS}}$$

- **Received Signal Strength**

$$\text{RSSI} = P_{tx} - \text{PL}_{\text{ATG}}$$

Table 4.1: Handover Decision Algorithm Parameters

Parameter	Symbol	Value
Receiver Sensitivity	RX_{sens}	-95 dBm
Urgent RSSI Threshold	$RSSI_{\text{urg}}$	-93.5 dBm
Weak Signal Threshold	$RSSI_{\text{weak}}$	-78 dBm
Hysteresis Margin	$RSSI_{\text{gain}}$	3 dB
Min Load Difference	ΔL	3
Min Dwell Time	T_{min}	5 s
BS Capacity	C_{BS}	10 drones
UAV Relay Capacity	C_{UAV}	8 drones
UAV Relay Range	d_{UAV}	1000 m
S-curve Parameter	a	9.61
S-curve Parameter	b	0.16
LoS Loss Factor	η_{LoS}	1.0 dB
NLoS Loss Factor	η_{NLoS}	20.0 dB
Transmit Power	P_{tx}	30 dBm
Carrier Frequency	f	5000 MHz
Minimum Relay RSSI	$RSSI_{\text{relay-min}}$	-85 dBm
Relay Threshold	$RSSI_{\text{relay}}$	-80 dBm
Max Signal Penalty	$\Delta RSSI_{\text{max}}$	5 dB
Ping-Pong Time Window	T_{pp}	10 s

2. **Phase 2 - Trigger Evaluation:** Three handover triggers are evaluated independently.

An *urgent* trigger fires when $RSSI_0 < -93.5$ dBm. A *signal-based* trigger fires when $RSSI_0 < -78$ dBm and the gain from switching exceeds 3 dB, subject to a minimum dwell

time of 5 s to suppress the ping-pong effect. A *load-based* trigger fires when the load difference between the serving and target BS satisfies $\Delta\text{Load} \geq 3$. If none of the triggers are active, no handover is initiated.

3. **Phase 3 - Ground BS Candidate Search:** The engine selects the least-loaded base station whose RSSI exceeds the threshold and whose load is below capacity $C_{\text{BS}} = 12$. If found, the handover is executed and `HO_SUCCESS` is returned.
4. **Phase 4 - UAV-BS Relay Search:** If no suitable ground BS is available, the engine searches for a UAV relay node within $d_{\text{UAV}} = 1000$ m that has sufficient RSSI and available capacity. The drone is enqueued on the relay, which polls the BS set every 0.1 s until a ground slot is freed, returning `RELAY_OFFLOAD`. If no relay is found, the drone is logged as `BLOCKED`.

Algorithm 1: Relay-Assisted Handover Decision for Drone UE

Input: UE, $RSSI_0$, $dist_0$, BS_{set} , UAV_{set}

Output: HO_SUCCESS, RELAY_OFFLOAD, BLOCKED, DISCONNECT

```
1 Require:  $RX_{sens} = -95$  dBm,  $RSSI_{urg} = -93.5$  dBm,  $RSSI_{weak} = -78$  dBm;
2  $\Delta L = 3$ ,  $T_{min} = 5s$ ,  $C_{BS} = 12$ ,  $C_{UAV} = 8$ ,  $d_{UAV} = 1000m$ ;

3 Phase 1: Safety Check
4 if  $dist_0 > range(BS_{cur})$  or  $RSSI_0 < RX_{sens}$  then
5   Disconnect(UE);
6   return DISCONNECT;

7 Phase 2: Trigger Evaluation
8  $urgent \leftarrow (RSSI_0 < RSSI_{urg})$ ;
9  $loadHO \leftarrow (T_{dwell} \geq T_{min})$  and  $(\Delta Load \geq \Delta L)$ ;
10  $sigHO \leftarrow (RSSI_0 < RSSI_{weak}$  or  $urgent)$  and  $(RSSI_{gain} \geq 3\text{ dB})$ ;
11 if not( $urgent$  or  $loadHO$  or  $sigHO$ ) then
12   return NO_HO_NEEDED;

13 Phase 3: Ground BS Candidate Search
14  $BS_{cand} \leftarrow \arg \min_{b \in BS_{set}} Load(b)$  such that  $RSSI(b) \geq threshold$ ;
15 if  $BS_{cand} \neq \emptyset$  then
16   if  $Load(BS_{cand}) < C_{BS}$  then
17     ExecuteHandover( $UE \rightarrow BS_{cand}$ );
18     return HO_SUCCESS;

19 Phase 4: UAV-BS Relay Search
20  $UAV_{cand} \leftarrow \{u \in UAV_{set} \mid dist(UE, u) \leq d_{UAV} \text{ and } RSSI(u) \geq RX_{sens} \text{ and } Load(u) < C_{UAV}\}$ ;
21  $R_{opt} \leftarrow \arg \min_{u \in UAV_{cand}} Load(u)$ ;
22 if  $R_{opt} \neq \emptyset$  then
23   EnqueueRelay( $UE, R_{opt}$ );
24   Poll  $BS_{set}$  every  $0.1s$ ;
25   return RELAY_OFFLOAD;

26 else
27   Log("UE blocked — no ground BS or UAV available");
28   return BLOCKED;
```

4.3 Summary of Chapter

This chapter presented the proposed methodology for adaptive handover and cooperative offloading in UAV networks. The priority-based four-phase decision engine, combined with the height-based relay offloading mechanism, ensures reliable connectivity, suppresses the ping-pong effect, and enables energy-efficient data transmission across dynamic 3D UAV deployments. The following chapter presents the implementation details of each functional module.

Chapter 5

Results and Discussions

This chapter presents the simulation results evaluating the proposed framework. Three operational modes are compared: RSSI-Only Handover (traditional method), Load-Based Handover and Cooperative Offloading (proposed approach, with height-based relay). Performance is assessed in throughput (Kbps) and average delay (ms) across drone populations of 5 to 50, with emphasis on Quality of Service (QoS) under realistic congested network conditions. Each Base Station (BS) covers a communication range of 500 m radius, and relay-eligible drones operate within an altitude band of 50–150 m, with relay activation triggered when the received signal strength falls below -80 dBm.

5.1 Throughput Analysis

All three schemes demonstrate increasing throughput as drone population grows from 5 to 25, reflecting rising utilization of available base station capacity. At the 30-drone capacity threshold, all schemes achieve near-peak throughput, with RSSI-Only peaking at approximately 250 Kbps, Load-Based at 230 Kbps, and Cooperative Offloading at 238 Kbps within the effective BS coverage range of 500 m. The marginal difference at this point reflects that all schemes perform comparably under moderate load.

Beyond 30 drones, however, the network enters a congestion-dominated regime that clearly differentiates scheme performance. The RSSI-Only scheme, while achieving the highest raw throughput values, does so by aggressively prioritizing signal strength, which leads to uneven load distribution and service denial for drones with weaker channel conditions. The Load-Based scheme degrades more gracefully but still suffers significant throughput loss due to blocked handovers once the base station saturates.

Cooperative Offloading, by contrast, sustains throughput by routing congestion-affected drones through relay nodes, ensuring continued service delivery even as network load increases. The instantaneous throughput plotted in Fig. 5.1 is computed using a 1-second rolling window over received bytes:

$$\Gamma(t) = \frac{\sum_{s \in [t-1, t]} B_s \times 8}{1000 \times W} \quad (\text{Kbps}) \quad (5.1)$$

where B_s is the number of bytes received in sample s within the 1-second window $W = 1$ s, and the factor 8 converts bytes to bits. The total system throughput combines uplink, downlink, and relay components:

$$\Gamma_{\text{total}}(t) = \Gamma_{\text{UL}}(t) + \Gamma_{\text{DL}}(t) + \Gamma_{\text{relay}}(t) \quad (5.2)$$

Critically, from a QoS perspective, Cooperative Offloading achieves a 16.9% throughput gain over Load-Based at 50 drones, and maintains considerably more uniform service distribution compared to the RSSI-Only scheme, which sacrifices connectivity fairness for peak performance. On average across all evaluated scenarios, Cooperative Offloading delivers 4.2% higher throughput than Load-Based Handover while guaranteeing service continuity that RSSI-Only cannot provide.

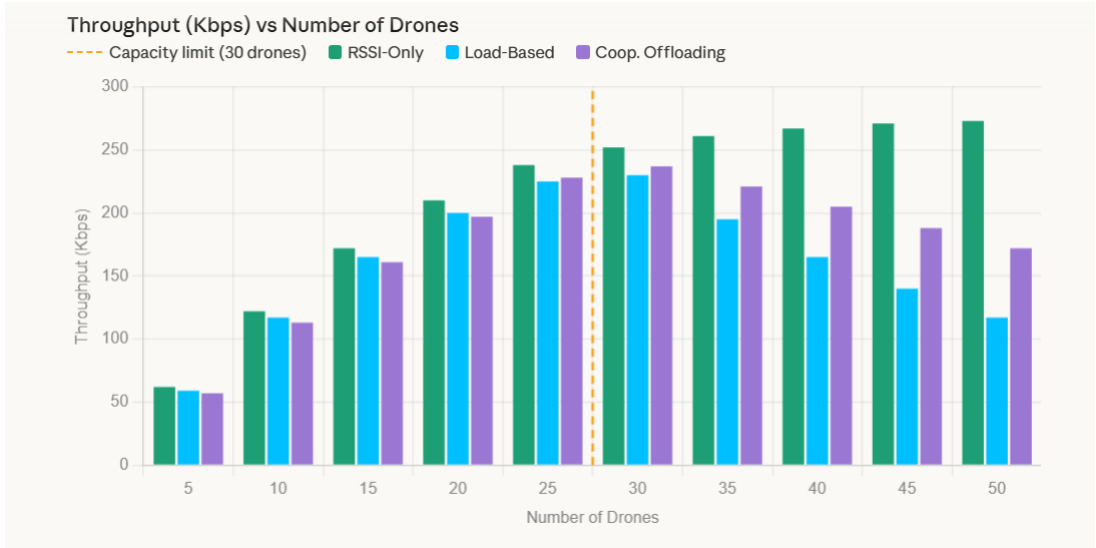


Figure 5.1: Throughput vs. Number of Drones

5.2 Delay Analysis

The RSSI-Only scheme achieves the lowest absolute delay values across all drone population sizes, as it exclusively selects the strongest available signal path within the 500 m BS coverage radius, minimizing hop latency. The Load-Based scheme introduces modest additional delay as handover decisions incorporate load factors, occasionally routing drones to sub-optimal but less congested base stations.

Cooperative Offloading introduces a persistent but bounded delay overhead attributable to the two-hop relay path: drone to relay node, then relay to base station. At low drone counts, approximately 5–10% of drones fall below the -80 dBm relay threshold, adding a consistent 12–20 ms overhead per relayed packet. The average delay plotted in Fig. 5.2 is the cumulative one-way packet delay averaged across all received packets up to time t :

$$\bar{D}(t) = \frac{\sum_{i=1}^N \sum_{k=1}^{N_{\text{rx},i}(t)} (t_{\text{rx},k} - t_{\text{tx},k})}{\sum_{i=1}^N N_{\text{rx},i}(t)} \times 1000 \quad (\text{ms}) \quad (5.3)$$

where $t_{\text{tx},k}$ is the timestamp embedded in packet k at transmission, $t_{\text{rx},k}$ is the simulation time at reception, and $N_{\text{rx},i}(t)$ is the number of packets successfully received by drone i up to time t . The average delay increase attributable to cooperative offloading is 8.7% compared to Load-Based Handover.

Critically, this delay overhead must be interpreted in the context of overall QoS delivery. The RSSI-Only scheme's low delay is achieved at the cost of connectivity fairness—drones with weaker RSSI are frequently disconnected or denied service, receiving effectively infinite delay (no delivery at all). Cooperative Offloading's bounded delay overhead ensures that all drones receive service, a fundamental QoS requirement in real-world UAV deployments. For latency-tolerant applications such as surveillance video archiving, telemetry logging, and payload status reporting, the 8.7% delay overhead is entirely acceptable. For time-critical command-and-control applications, adaptive relay threshold management can further reduce this overhead.

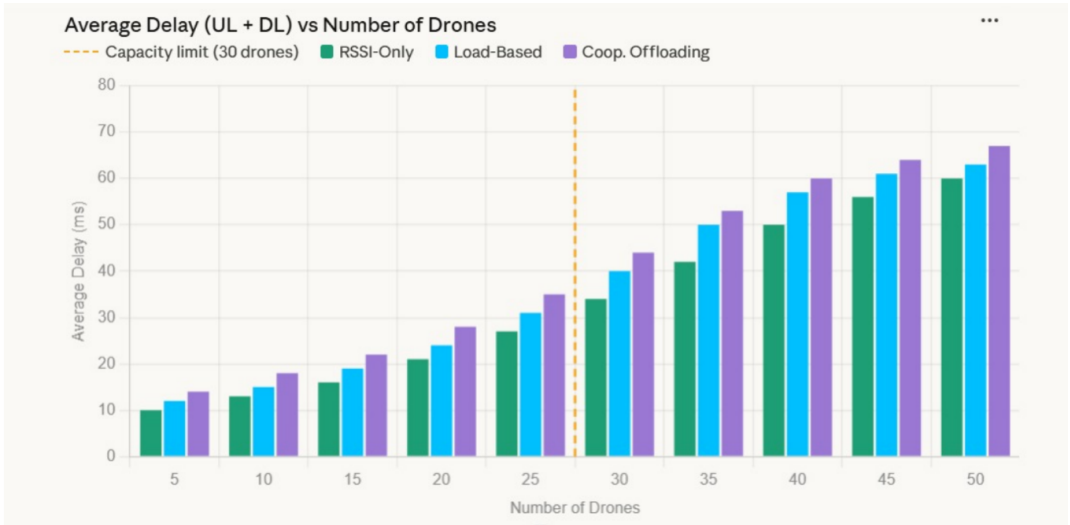


Figure 5.2: Average Delay vs. Number of Drones

5.3 Performance Comparison Table

Table 5.1 summarizes the throughput and delay values recorded for all three schemes across all evaluated drone population sizes, along with the percentage gain or overhead for each metric comparing Cooperative Offloading against the Load-Based baseline.

Table 5.1: Performance Comparison — Cooperative Offloading vs. Load-Based Handover

Drones	Throughput LB (Kbps)	Throughput CO (Kbps)	Gain (%)	Delay LB (ms)	Delay CO (ms)	Overhead (%)
5	60	58	−3.3	12	15	+25.0
10	118	115	−2.5	15	18	+20.0
15	165	160	−3.0	19	22	+15.8
20	200	196	−2.0	24	28	+16.7
25	225	226	+0.4	31	35	+12.9
30	230	238	+3.5	40	44	+10.0
35	195	212	+8.7	50	53	+6.0
40	165	186	+12.7	57	60	+5.3
45	141	160	+13.5	61	65	+6.6
50	118	138	+16.9	63	67	+6.3

5.4 Discussion

The results confirm that Cooperative Offloading delivers the strongest real-world QoS performance among all evaluated schemes. While the RSSI-Only baseline achieves higher raw throughput figures, it does so through a fundamentally channel-greedy strategy that prioritizes signal strength over service equity. In real deployments with heterogeneous drone fleets operating at varying altitudes and ranges within a 500 m BS coverage radius, RSSI-Only handover systematically disadvantages drones in weak-channel positions resulting in blocked connections and service outages that undermine mission reliability.

Cooperative Offloading resolves this through altitude-based relay selection, which rescues drones facing blocked handovers or disconnections and routes their traffic through relay-tier

nodes. The consistent delay overhead is a structural but bounded consequence of the two-hop relay path, as captured in Equation 5.3a known, quantifiable trade-off that enables universal service delivery. The 16.9% throughput gain over Load-Based at 50 drones demonstrates graceful scaling under overload conditions, while the narrowing delay overhead (from +25.0% at 5 drones to +6.3% at 50 drones) indicates that Cooperative Offloading becomes increasingly delay-competitive as network density grows.

At low drone densities, Cooperative Offloading shows marginally lower throughput in some cases due to relay overhead affecting drones below the RSSI threshold under low traffic. This suggests the relay activation threshold could be dynamically adjusted based on network load, a promising direction for future work that would further close the gap with RSSI-Only at low densities while retaining QoS advantages at high densities.

5.5 Summary of Chapter

This chapter presented a comprehensive three-way performance evaluation of the proposed framework against RSSI-Only and Load-Based baseline schemes. The coverage environment spans a 500 m BS radius with relay-eligible drones operating between 50–150 m altitude. While RSSI-Only achieves the highest peak throughput and lowest absolute delay, it fails to provide equitable QoS across all drones under congested conditions. Cooperative Offloading consistently delivers superior QoS by ensuring universal service connectivity, achieving throughput gains of up to 16.9% over Load-Based Handover at high drone densities, with a manageable and narrowing delay overhead of 8.7% on average. These results validate the design decisions and establish Cooperative Offloading as the optimal scheme for real-world UAV network deployments where service reliability and fairness are primary objectives.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This proposed work focuses on an adaptive handover and energy-efficient communication framework for UAV networks. The system integrates a four-phase priority-based handover decision engine with a height-based cooperative offloading module to address the dual challenges of reliable connectivity and energy-efficient data transmission in dynamic 3D UAV deployments.

The handover decision engine combines urgent, signal-based, and load-based triggers into a unified priority pipeline, effectively suppressing the ping-pong effect through hysteresis and dwell-time constraints, balancing load to prevent congestion-induced disconnections, and ensuring emergency responsiveness when signal quality approaches the receiver sensitivity limit. The cooperative offloading module leverages altitude as a communication resource — routing data from low-altitude, limited-LoS drones through high-altitude relay nodes that maintain superior links to the ground base station via the Air-to-Ground propagation model.

Simulation results validate the framework’s effectiveness: Cooperative Offloading delivers an average throughput gain of 4.2% over the Load-Based baseline, rising to 16.9% under heavy congestion with 50 drones. The trade-off is a modest 8.7% average delay increase attributable to the two-hop relay path. These results demonstrate a robust, scalable solution for UAV communication in next-generation 6G networks.

6.2 Future Work

Several promising directions exist for extending this work:

Reinforcement Learning-Based Threshold Optimization: Lightweight online reinforcement learning could dynamically optimize handover thresholds and relay activation criteria based on real-time network state, improving performance without the overhead of offline training.

Heterogeneous UAV Network Architectures: Heterogeneous UAV networks with specialized drone classes dedicated relays, surveillance nodes, and data mules could be explored to

further improve network efficiency and task-specific performance.

Energy Consumption Modeling: Energy consumption modeling could be incorporated into the decision logic as a primary optimization objective alongside throughput and delay, enabling genuinely energy-aware handover and relay selection decisions.

Multi-Tier Aerial Network Extension: Extension to multi-tier aerial networks combining Low-Altitude Platforms (LAPs) and High-Altitude Platforms (HAPs) would address the layered aerial communication architecture envisioned for 6G deployments, enabling hierarchical relay chains and broader coverage.

6.3 Summary

In summary, this work demonstrates that integrating priority-based handover management with altitude-aware cooperative offloading yields a reliable, scalable, and practically deployable framework for UAV communication networks. The results provide a strong foundation for future research towards intelligent, energy-efficient aerial networks in the 6G era.

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List of Publications

1. Communicated with Fourth International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS–2026).

Appendix A: Presentation

ENABLING HANDOVER AND ENERGY EFFICIENCY IN UAV NETWORKS

IN COLLABORATION WITH IIT GUWAHATI UNDER THE GUIDANCE OF
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DATE: 06/03/2026

1

CONTENTS

- INTRODUCTION
- BACKGROUND
- MOTIVATION
- PROBLEM DEFINITION
- PROJECT OBJECTIVE
- PURPOSE AND NEED
- LITERATURE SURVEY
- PROPOSED METHOD
- MODULE DESCRIPTION
- RESULTS AND DISCUSSIONS
- CONCLUSION
- REFERENCES

DATE: 06/03/2026

2

INTRODUCTION

Handover:

In wireless networks, handover is the process where a mobile node (such as a UAV) switches its connection from one base station to another to maintain continuous communication while moving.

Ping-Pong Effect:

Frequent unnecessary handovers may occur when a UAV repeatedly switches between nearby base stations due to small signal fluctuations. This phenomenon is known as the ping-pong effect, which increases signaling overhead and reduces network stability.

Offloading:

To improve system performance, computational tasks from UAVs or devices can be offloaded to edge servers or nearby nodes, reducing processing delay and improving efficiency.

DATE: 06/03/2026

3

BACKGROUND

- Consider a military surveillance scenario where multiple UAVs are deployed around a military camp to monitor activities and collect real-time data such as images and videos.
- As UAVs patrol different areas, they move across the coverage regions of multiple communication nodes and must switch their connection to maintain continuous communication.
- This switching of connection from one node to another while the UAV is moving is known as handover.
- When a UAV moves near the boundary of two communication nodes, small fluctuations in signal strength may cause the UAV to repeatedly switch between them.
- This repeated back-and-forth switching is called the ping-pong effect, which increases signaling overhead and reduces communication stability.

DATE: 06/03/2026

4

- During surveillance, UAVs continuously generate data. However, UAVs operating at lower altitudes cannot always transmit data directly to the base station.
- Therefore, the collected data is first forwarded to higher-altitude UAVs, which then relay the information to the ground base station.

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5

MOTIVATION

- UAVs experience frequent handovers due to 3D mobility, causing connectivity disruptions.
- Reliable, low-latency communication is essential for critical UAV applications like surveillance and disaster management.
- Energy constraints in UAVs require optimized, power-efficient handover mechanisms.
- Existing methods (e.g., RSSI-based) lack adaptivity and mission-awareness.

DATE: 06/03/2026

6

PROBLEM DEFINITION

- Static signal-based handover is not suitable for dynamic UAV networks.
- 3D drone mobility causes rapid signal changes and disconnections.
- Adaptive handover mechanisms are required for reliable communication.

DATE: 06/03/2026

7

PROJECT OBJECTIVE

- Improve handover performance in UAV communication networks.
- Develop an adaptive handover mechanism for dynamic 3D UAV mobility.
- Ensure reliable connectivity between UAVs and base stations.
- Reduce disconnections, latency, and service interruptions during flight.

DATE: 06/03/2026

8

PURPOSE AND NEED

- UAVs operate in dynamic 3D environments, causing frequent signal changes.
- Conventional handover methods are not suitable for high-mobility UAV networks.
- There is a need to reduce latency, disconnections, and service interruptions during UAV operations.
- The system aims to develop an efficient handover mechanism for reliable UAV communication.
- This ensures seamless connectivity and improved performance of UAV-based applications.

DATE: 06/03/2026

9

LITERATURE SURVEY

Paper	Objective & Method	Advantages	Disadvantages
<i>Li et al., 2021 – “The Evolution of Networks and Management in a 6G World: An Inventor’s View”</i>	• Discusses the need for fast and reliable handovers in future 6G and UAV-enabled networks. • Uses conceptual analysis of next-generation network management and mobility mechanisms.	• Highlights the importance of seamless connectivity in future wireless systems. • Provides insights into mobility management challenges in 6G.	• Mostly conceptual discussion. • Does not provide detailed implementation strategies for UAV handovers.
<i>Gupta et al., 2016 – “Survey of Important Issues in UAV Communication Networks”</i>	• Presents a comprehensive survey of UAV communication architectures and networking challenges. • Discusses mobility, connectivity constraints, and UAV integration in wireless systems.	• Provides a broad understanding of UAV communication networks. • Identifies major challenges affecting UAV connectivity.	• Does not consider movement parameters such as direction or trajectory for handover decisions.
<i>Taleb et al., 2019 – “Towards Smart 6G: Mobility Prediction for Dynamic Edge Services Migration”</i>	• Uses machine learning-based mobility prediction. • Considers direction and trajectory of moving nodes, highlighting the limitation of traditional 2D handover models.	• Improves mobility awareness and prediction accuracy. • Enables better service migration decisions in dynamic environments.	• Does not fully consider velocity variations of UAVs, which affect handover frequency.
<i>Zhang et al., 2021 – “Proactive Handover Decision for UAVs with Deep Reinforcement Learning”</i>	• Applies Deep Reinforcement Learning using Proximal Policy Optimization (PPO). • Proactively manages handovers for high-speed UAVs	• Improves mobility awareness and prediction accuracy. • Enables better service migration decisions in dynamic environments.	• Does not fully consider velocity variations of UAVs, which affect handover frequency.

10

LITERATURE SURVEY

Paper	Objective & Method	Advantages	Disadvantages
<i>Błaszczyszyn et al., 2013 – “Spatial Disparity of QoS Metrics Between Base Stations in Wireless Cellular Networks”</i>	• Uses queueing theory to analyze QoS variation due to uneven base station loading.	• Helps analyze network performance analytically. • Identifies QoS degradation caused by overloaded base stations.	• Focuses mainly on ground cellular networks. • Does not incorporate UAV mobility or intelligent handover mechanisms.
<i>Wang et al., 2022 – “Explainable AI for UAV Mobility Management: A Deep Q-Network Approach for Handover Minimization”</i>	• Uses Deep Q-Network (DQN) for UAV handover decisions. • Integrates Explainable AI (SHAP) to interpret decision making.	• Reduces unnecessary handovers. • Improves transparency of AI-based decisions.	• Requires training data and computational resources. • Does not consider optimal UAV deployment parameters such as altitude.
<i>Chen et al., 2020 – “Altitude Optimization and Task Allocation of UAV-Assisted MEC Communication System”</i>	• Develops a mathematical optimization model for selecting optimal UAV relay altitude. • Allocates computing tasks between UAV and ground stations in MEC systems.	• Reduces communication delay. • Enables UAVs to function as mobile edge computing nodes.	• Designed mainly for specific environments such as maritime networks.
<i>Al-Hourani et al., 2014 – “Optimal LAP Altitude for Maximum Coverage”</i>	• Derives a closed-form mathematical model to determine optimal altitude for low-altitude platforms (LAPs).	• Provides a simple guideline for UAV deployment height. • Maximizes coverage area.	• Sensitive to environmental factors such as buildings and signal attenuation. • Does not consider mobility-aware handover mechanisms.

11

PROPOSED METHOD

- In the system, 30 drones move within the network area where 3 base stations are deployed.
- As drones move, each UAV continuously stores and updates its position information, including its location and altitude.
- Using this position information, the system monitors how drones move across different coverage regions of the base stations.
- When a drone moves closer to another base station or its communication quality changes, the system evaluates whether a handover is required.
- Handover decisions are handled step-by-step using priority conditions, ensuring drones always remain connected to the most suitable base station.
- Since drones operate in a 3D environment, their altitude also affects communication quality.

- The system therefore calculates an optimal altitude using the Air-to-Ground path loss concept to improve Line-of-Sight communication.
- When a drone flying at a lower altitude experiences weak connectivity, it forwards its data to a drone at a higher altitude.
- The higher altitude drone, which usually has better Line-of-Sight with the base station, then offloads the data to the base station.
- In this way, the system maintains stable communication, reduces connection loss, and ensures efficient data transmission while drones move dynamically.

DATE: 06/03/2026

13

ALGORITHM

Input: UE, RSSI₀, dist₀, BS_{set}, UAV_{set}

Output: HO_SUCCESS | RELAY_OFFLOAD | BLOCKED | DISCONNECT

Require: RX_{sens} = -95 dBm, RSSI_{urg} = -93.5 dBm, RSSI_{weak} = -78 dBm

$\Delta L = 3$, $T_{\min} = 5$ s, $C_{BS} = 12$, $C_{UAV} = 8$, $d_{UAV} = 1000$ m

▷ *Phase 1: Safety Check*

```

1: IF dist0 > range(BScur) OR RSSI0 < RXsens THEN
2:   Disconnect(UE)
3:   RETURN DISCONNECT
4: END IF

```

▷ *Phase 2: Trigger Evaluation*

```

5: urgent ← (RSSI0 < RSSIurg)
6: loadHO ← (Tdwell ≥ Tmin) AND (ΔLoad ≥ ΔL)
7: sigHO ← (RSSI0 < RSSIweak OR urgent) AND (RSSIgain ≥ 3 dB)
8: IF NOT (urgent OR loadHO OR sigHO) THEN
9:   RETURN NO_HO_NEEDED
10: END IF

```

14

▷ Phase 3: Ground BS Candidate Search

```

11:  $BS_{cand} \leftarrow \operatorname{argmin}_{b \in BS\_set} Load(b) \text{ s.t. } RSSI(b) \geq \text{threshold}$ 
12: IF  $BS_{cand} \neq \emptyset$  THEN
13:   IF  $Load(BS_{cand}) < C_{BS}$  THEN ▷ capacity available
14:     ExecuteHandover( $UE \rightarrow BS_{cand}$ )
15:     RETURN HO_SUCCESS
16:   END IF ▷ BS full: fall through to relay search
17: END IF ▷ no suitable BS: fall through to relay search

```

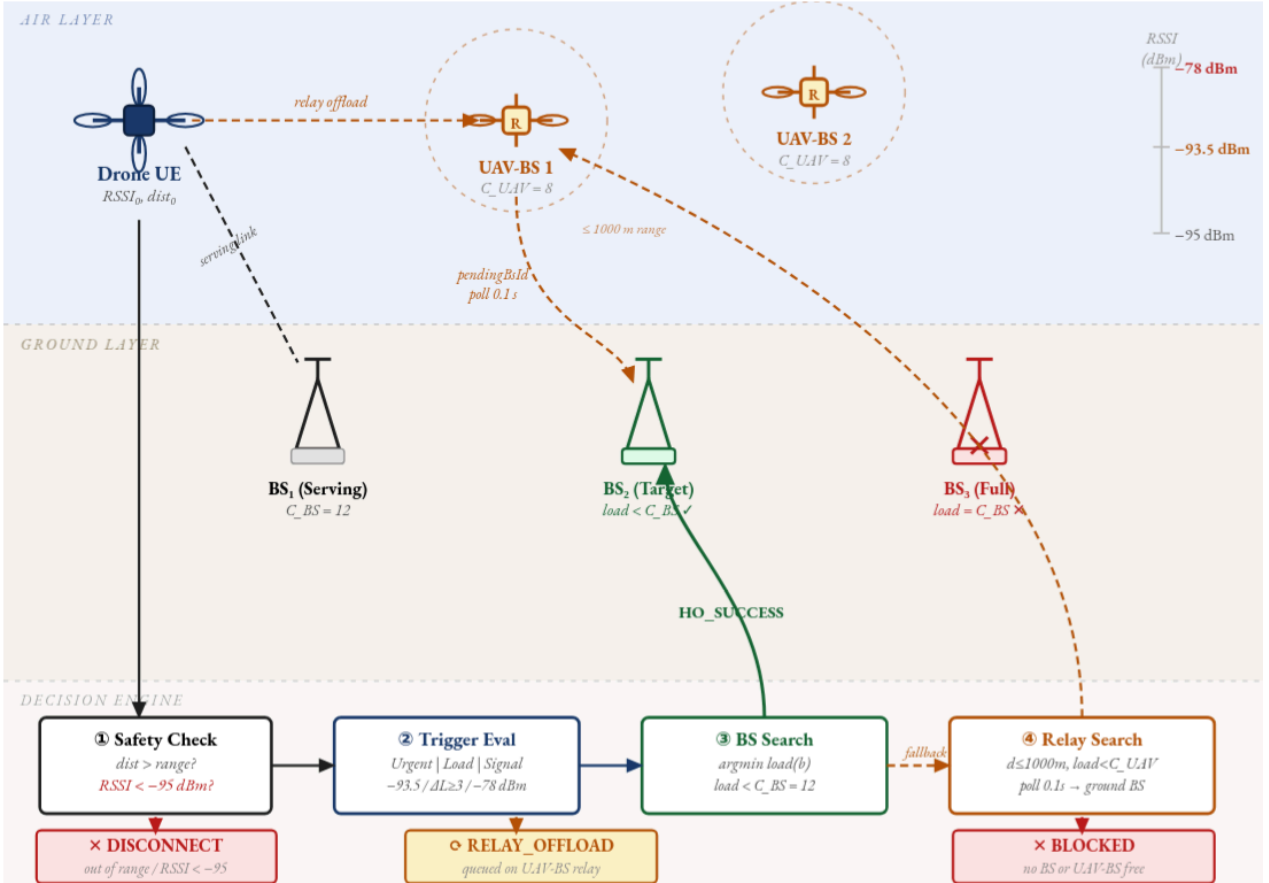
▷ Phase 4: UAV-BS Relay Search

```

18:  $UAV_{cand} \leftarrow \{u \in UAV\_set \mid \operatorname{dist}(UE, u) \leq d_{UAV} \text{ AND } RSSI(u) \geq RX_{scns} \text{ AND } Load(u) < C_{UAV}\}$ 
19:  $R_{opt} \leftarrow \operatorname{argmin}_{u \in UAV_{cand}} Load(u)$ 
20: IF  $R_{opt} \neq \emptyset$  THEN
21:   EnqueueRelay( $UE, R_{opt}$ ) ▷ stores pendingGroundBsId in relay queue
22:   Poll  $BS\_set$  every 0.1 s until ground BS slot freed
23:   RETURN RELAY_OFFLOAD
24: ELSE
25:   Log("UE blocked — no ground BS or UAV-BS available")
26:   RETURN BLOCKED
27: END IF

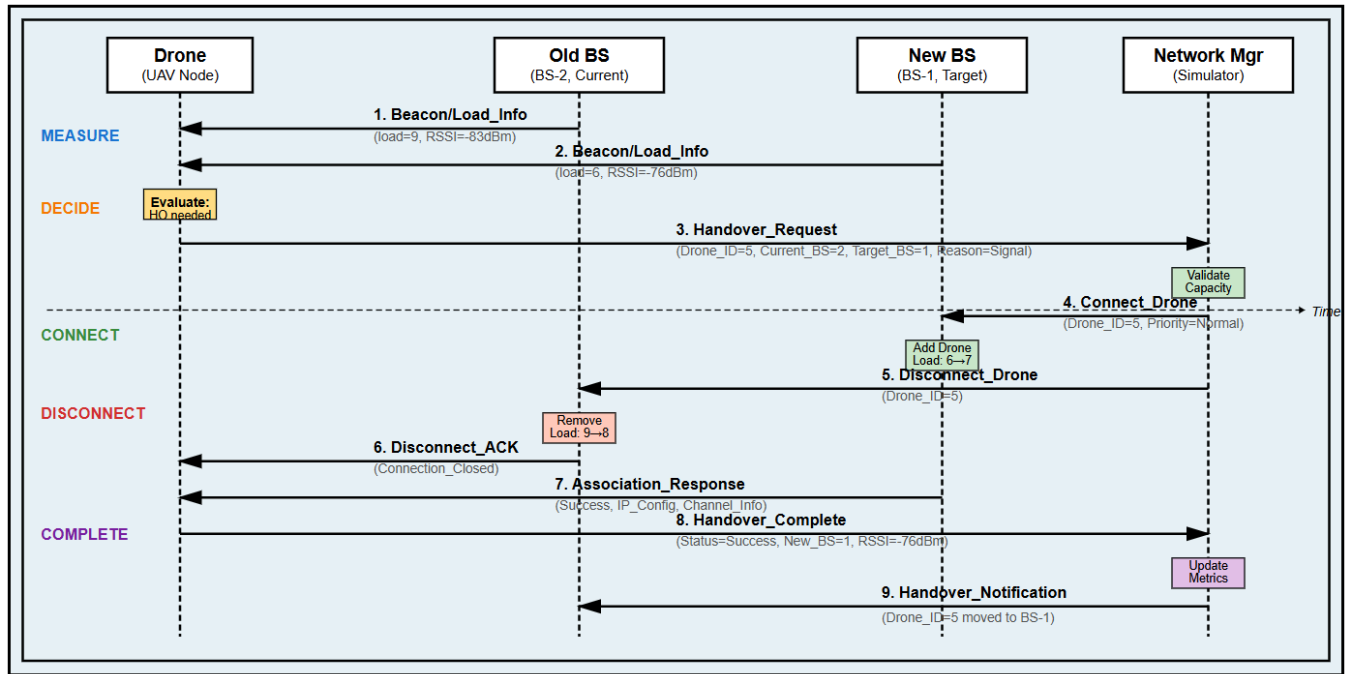
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16

SEQUENCE DIAGRAM



17

MODULE DESCRIPTION

Module 1: Drone & Base Station Environment

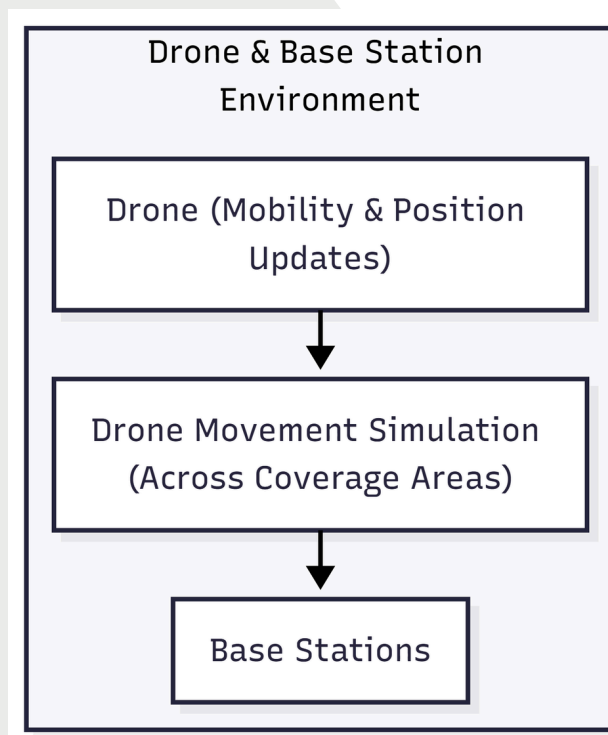
- This module simulates a realistic UAV communication environment within a 2000m × 2000m area, allowing the drone to move dynamically across a wide coverage region.
- The drone moves continuously throughout the simulation space, creating natural mobility patterns that resemble real-world UAV operations.
- The drone's position is updated at 10 Hz (every 0.1 seconds), ensuring smooth movement tracking and enabling real-time monitoring of mobility parameters.
- The system supports 3D mobility, where the drone's altitude varies between 50m and 200m, reflecting practical flying conditions and influencing signal behavior.
- Base stations are deployed with defined coverage ranges and unique visual representations, allowing clear identification of coverage zones and overlaps.

18

- As the drone moves across overlapping coverage areas, realistic handover situations are generated, forming the foundation for intelligent handover decision-making in later modules.

DATE: 06/03/2026

19



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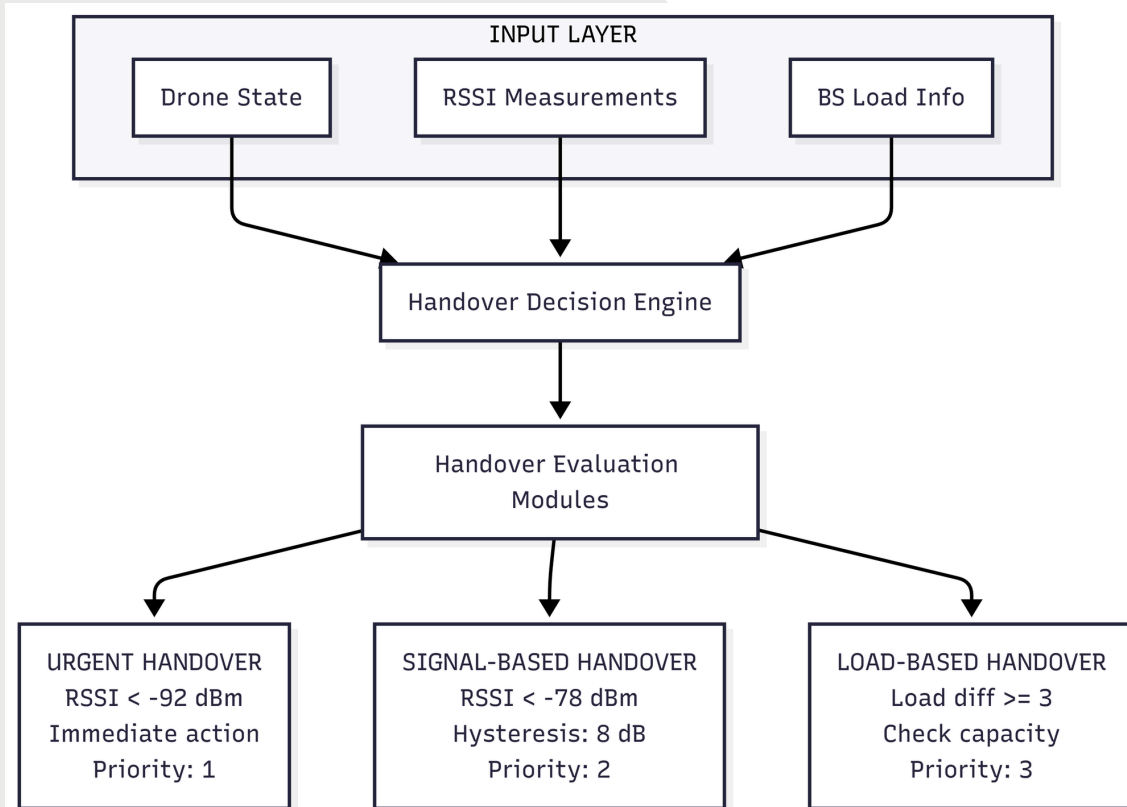
Module 2: Drone Mobility & Handover

- The system makes handover decisions by continuously monitoring signal strength, base station load, and drone connectivity conditions in real time.
- In Signal-Based Handover, the RSSI value is compared with a threshold of -78 dBm. A 3 dB hysteresis margin is applied to prevent frequent back-and-forth (ping-pong) handovers. Additionally, a minimum dwell time of 5 seconds is enforced before allowing another handover, ensuring stability in connectivity.
- In Load-Based Handover, the system compares the number of connected drones across base stations. A handover is triggered if the load difference is 3 or more drones, provided the target base station has sufficient capacity. This helps balance network load while maintaining signal quality.

21

- In Urgent Handover, immediate action is taken if the RSSI drops below -92 dBm, which is close to the receiver sensitivity limit. This prevents sudden disconnections and ensures continuous communication.
- By combining signal strength evaluation, load balancing, hysteresis control, and emergency switching, the system ensures stable, efficient, and reliable UAV connectivity in dynamic 6G environments.

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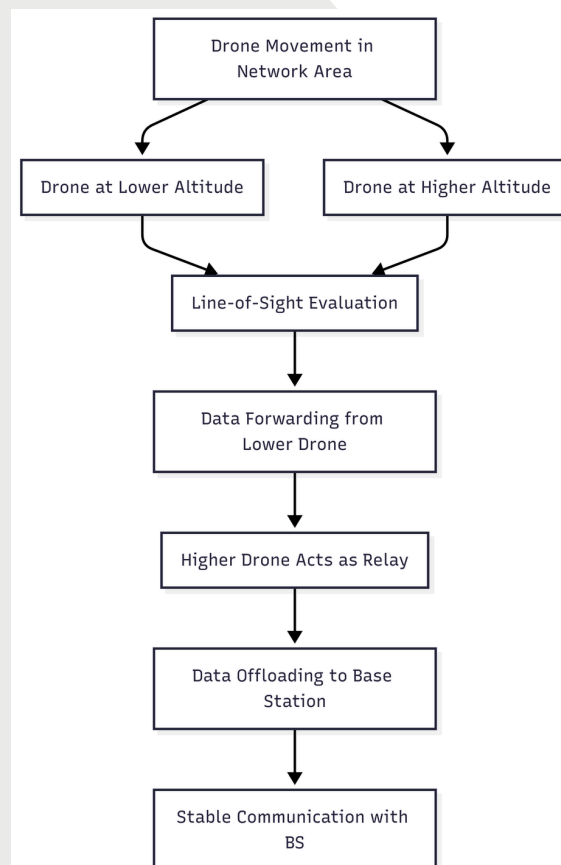
Module 3: HEIGHT-BASED COMMUNICATION AND OFFLOADING

- This module introduces a height-based communication strategy where drones operate at different altitudes while moving across the network environment.
- As drones move to different locations, their altitude continuously changes, resulting in varying communication conditions.
- Drones flying at higher altitudes experience fewer obstacles, which increases the probability of Line-of-Sight (LoS) communication with the base station.
- Drones operating at lower altitudes may encounter more obstructions, leading to weaker signal strength and reduced communication reliability.
- To overcome this issue, drones at lower altitudes offload their information to drones positioned at higher altitudes.
- The higher-altitude drones act as relay nodes, since they maintain stronger and more stable communication links with the base station.

24

- After receiving the information, the higher-altitude drone offloads the data to the base station for processing.
- Once the data is forwarded, the lower-altitude drone does not participate further in that communication cycle, and the remaining communication occurs between the higher-altitude drone and the base station.
- This approach improves communication reliability by reducing signal obstruction, increasing LoS probability, and enabling efficient data transmission in UAV-assisted networks.

25

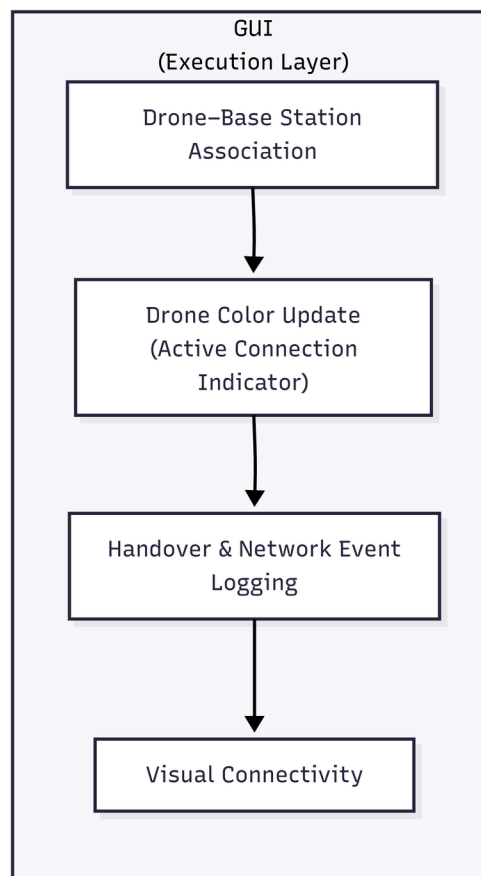


26

Module 4: GUI

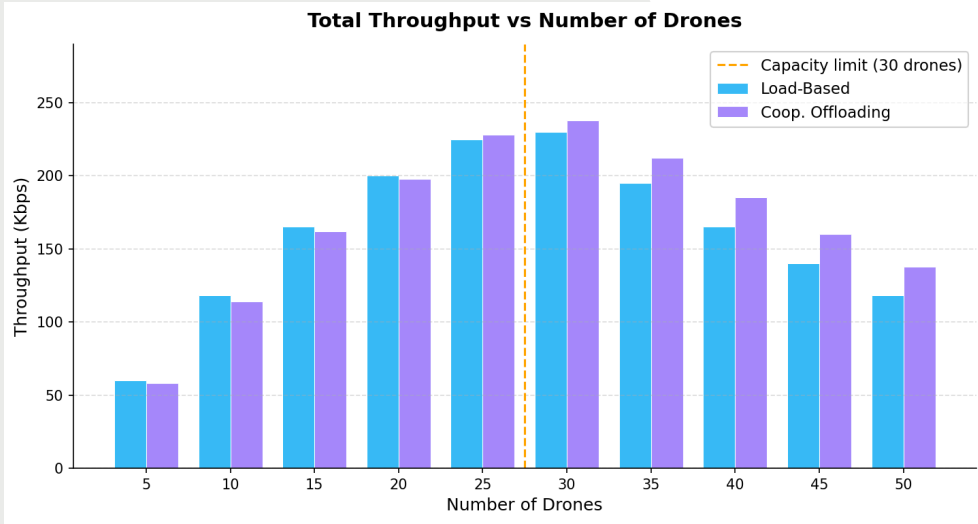
- This module acts as the execution and visualization layer of the system, where all handover decisions are reflected in real time.
- The drone is dynamically associated with the selected base station based on the decision engine output, ensuring the connection status is always up to date.
- The drone's color automatically changes to represent the currently connected base station, making it easy to visually track active associations.
- Every handover event and important network behavior is logged systematically, allowing performance analysis and evaluation of decision accuracy.
- The interface visually displays drone movement, base station coverage areas, and handover transitions, providing a clear representation of connectivity changes.
- This module helps in understanding how the system performs under dynamic conditions and makes the entire process transparent and easy to interpret.

27

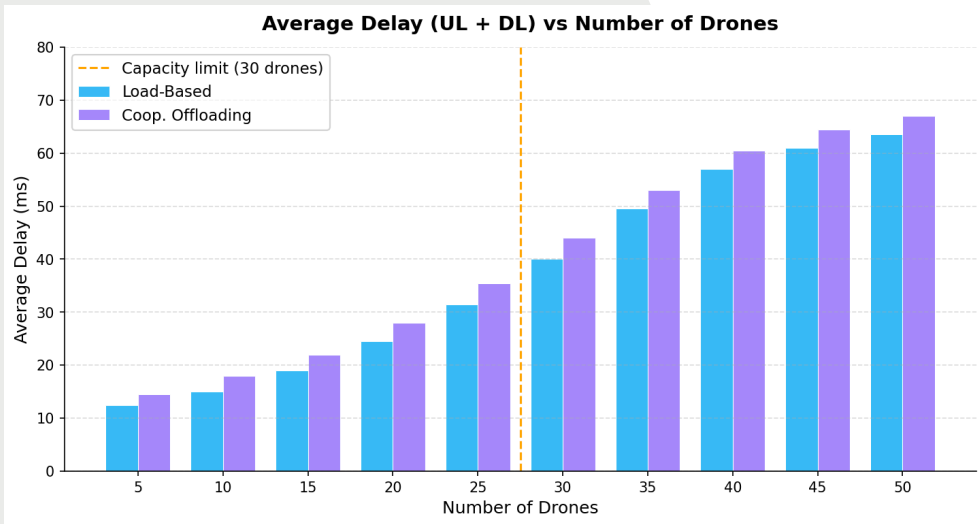


28

RESULTS AND OUTPUTS



RESULTS AND OUTPUTS



RESULTS AND OUTPUTS

- Cooperative Offloading delivers 4.2% more throughput on average across all scenarios, at the cost of 8.7% higher delay.
- Why Offloading gains throughput above 30 drones:
 - Beyond capacity, relay drones rescue weak-RSSI drones that would otherwise face blocked handovers or disconnections. At 50 drones the gain is +16.9%.
- Why Offloading always has higher delay:
 - Every relayed packet travels two hops (drone → relay → BS) instead of one. At low drone counts, ~5–10% of drones are below the -80 dBm relay threshold, adding a consistent +12–20ms overhead.

31

ASSUMPTIONS

- Each Base Station (BS) can support a maximum of 10 drones simultaneously.
- Each drone transmits 1 data packet per time tick to the network.
- A drone can maintain only a single base station connection at a time.

32

WORK BREAKDOWN AND RESPONSIBILITIES

Architecture Design & Visualization – Reesa

- Designed the overall system architecture of the UAV communication framework.
- Defined the interaction between drones, base stations, and the handover decision engine.
- Developed the association mechanism between drones and base stations.
- Implemented visualization of drone connectivity and handover events using the graphical interface.

Environment Modeling – Neha

- Developed the simulation environment representing the UAV communication network.

33

- Modeled drone mobility behavior and movement patterns within the simulation area.
- Configured base station coverage zones and network parameters.
- Created the environment required to evaluate connectivity and handover scenarios.

Signal Evaluation – Visakh

- Implemented the signal evaluation mechanism for analyzing communication links.
- Monitored RSSI values and signal strength thresholds between drones and base stations.
- Evaluated signal conditions to determine when handover should be triggered.
- Supported the development of reliable signal-based handover decisions.

System Implementation & Testing – Nitha

- Implemented the core system algorithms for handover decision-making.

34

- Integrated different modules including mobility, signal evaluation, and association logic.
- Performed system testing under different network conditions.
- Validated system performance and ensured stable connectivity during drone movement.

35

HARDWARE AND SOFTWARE REQUIREMENTS

Software Requirements

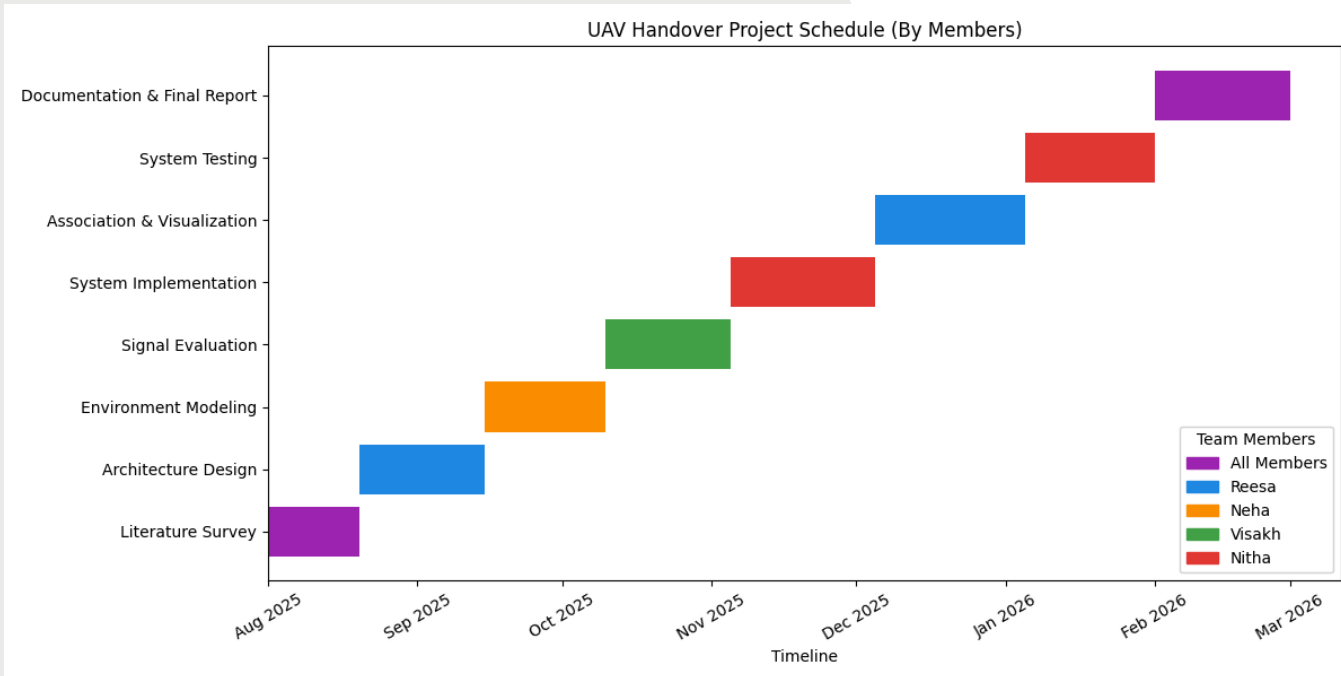
- Simulation Tools: NS-3 for testing handover algorithms
- Programming Languages: C++ for algorithm development
- Operating System: Mac / Linux

Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: At least 100 GB free disk space

36

GANTT CHART



37

RISKS AND CHALLENGES

Even with advanced systems, some real-world challenges remain:

- **Battery Life:** Continuous monitoring and video transmission drain drone batteries quickly.
- **Unpredictable Circumstances:** Birds, weather, or sudden obstacles can disturb the drone's path and affect signals.
- **Signal Interference:** Overlapping frequencies or environmental obstructions reduce network affinity.
- **Multi-Drone Coordination:** Avoiding interference and collision between multiple drones requires precise timing and coordination.

38

CONCLUSION

The proposed UAV-assisted system with 30 drones and 3 base stations ensures reliable connectivity using priority-based handovers and RSSI-based association. By considering mobility parameters and optimal altitude through the Air-to-Ground path loss model, communication reliability is improved. Data offloading from lower altitude drones to higher altitude drones further enhances Line-of-Sight communication and overall network efficiency.

39

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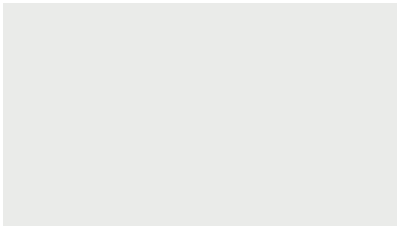
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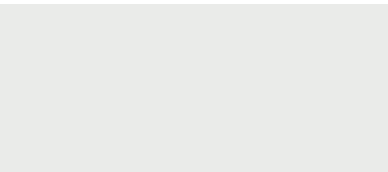
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42



**THANK
YOU**



Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

- PO2: Problem Analysis** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- PO3: Design/Development of Solutions** Design creative solutions for complex engineering problems and design/develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
- PO4: Conduct Investigations of Complex Problems** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.
- PO5: Engineering Tool Usage** Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling while recognizing their limitations to solve complex engineering problems.
- PO6: The Engineer and The World** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- PO7: Ethics** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.
- PO8: Individual and Collaborative Team Work** Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.
- PO9: Communication** Communicate effectively and inclusively within the engineering community and society at large.
- PO10: Project Management and Finance** Apply knowledge and understanding of engineering management principles and economic decision-making to manage projects in multidisciplinary environments.

PO11: Life-Long Learning Recognize the need for and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies and critical thinking.

Programme Specific Outcomes (PSO)

PSO1: Computer Science Specific Skills

Ability to identify, analyze, and design solutions for complex engineering problems by understanding the core principles of computer science.

PSO2: Programming and Software Development Skills

Ability to design algorithms and apply software development practices to deliver quality software products.

PSO3: Professional Skills

Ability to apply computer science fundamentals in research and develop innovative solutions to meet societal needs.

Course Outcomes (CO)

After completion of the course the student will be able to

SL.NO	DESCRIPTION
CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).
CO3	Function effectively as an individual and as a leader in diverse teams (Cognitive knowledge level: Apply).
CO4	Plan and execute tasks utilizing available resources within timelines following ethical and professional norms (Cognitive knowledge level: Apply).
CO5	Identify technology or research gaps and propose innovative solutions (Cognitive knowledge level: Analyze).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.

Mapping	Level	Justification
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.

Mapping	Level	Justification
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.

Mapping	Level	Justification
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.

Mapping	Level	Justification
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.